Root data, Weyl chambers and the Bruhat decomposition

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Codex, Ultra setting

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Course sources: https://github.com/KokunoYumeto/open-math-courses/tree/main/docs/courses/AG-RG.

Contents

4 Root data, Weyl chambers and the Bruhat decomposition **1**

1. The integral data and their reflections 1
2. The root datum of a reductive group 3
3. Reflections and chambers 6
4. Borel subgroups are positive systems 11
5. Coordinates on the unipotent radicals 14
6. The Bruhat cells from a projective limit 17
7. Central quotients and their root coordinates 20
8. Splittings and subgroup constructions in families 22
9. Exercises 27

Solutions 27

References and prerequisites 28

Lesson 4

Root data, Weyl chambers and the Bruhat decomposition

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A root records how a torus moves a one-dimensional subgroup. Its coroot records the diagonal direction in the associated rank-one group. Keeping both, together with the integral character and cocharacter lattices, distinguishes groups that have the same root system. We first assemble these data, then use cocharacters to describe Borel subgroups and the Bruhat cells. The last part explains which of these constructions survive over an arbitrary base.

The preceding lesson supplies root groups, their canonical line-bundle parametrizations, coroots, and the limit subgroups $U_G(\lambda)$ and $P_G(\lambda)$. For the field arguments we also use the geometric Borel quotient and the simply transitive action of $N_G(T)/T$ on the Borels containing T, proved in the first two lessons. We prove the identification of this normalizer quotient with the reflection group here; Theorem 4.2 then proves arbitrary-base finiteness using this field calculation.

1. The integral data and their reflections

A **root datum** consists of finite free lattices X and $X^\vee$, a perfect pairing

$$\langle\ ,\ \rangle : X \times X^\vee \longrightarrow \mathbf{Z},$$

finite subsets $\Phi \subset X \setminus \{0\}$ and $\Phi^\vee \subset X^\vee \setminus \{0\}$, and a bijection $\alpha \mapsto \alpha^\vee$ between them. Its axioms are

$$\langle \alpha, \alpha^\vee \rangle = 2, \qquad s_\alpha(\Phi) = \Phi, \qquad s_\alpha^\vee(\Phi^\vee) = \Phi^\vee,$$

where

$$s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha, \qquad s_\alpha^\vee(y) = y - \langle \alpha, y \rangle \alpha^\vee. \tag{1.1}$$

We require the bijection to respect these reflections. The reflections square to the identity, are dual under the pairing, and send α and $\alpha^\vee$ to their negatives. These assertions follow by substituting $\langle \alpha, \alpha^\vee \rangle = 2$ in (1.1). A datum is **reduced** if $\Phi \cap \mathbf{Q}\alpha = \{\alpha, -\alpha\}$ for every root. The empty root set is allowed: it describes a torus. Roots need not span $X_{\mathbf{Q}}$.

The **Weyl group** $W(\mathcal{R})$ is the subgroup of $\mathrm{Aut}(X)$ generated by the s_α. We prove finiteness without assuming away the central directions. Put $V = \mathbf{Q}\Phi$. The image of W on V is finite, since

it permutes the finite spanning set Φ. Choose a rational basis of V, give its real extension the sum-of-squares inner product, and average that form over this finite image. The average is positive definite because every summand is nonnegative and a nonzero vector has positive squared length under every invertible group element; relabelling the finite sum proves invariance. Each $s_\alpha|_V$ is an orthogonal reflection: its minus-one line is $\mathbf{R}\alpha$ and its fixed hyperplane is $\ker\langle\ ,\alpha^\vee\rangle$. The two eigenspaces are orthogonal, since invariance gives $(a,u)=(-a,u)=-(a,u)$. Decomposition along that line therefore gives

$$\langle v,\alpha^\vee\rangle=\frac{2(v,\alpha)}{(\alpha,\alpha)}\quad(v\in V). \tag{1.2}$$

These restricted coroot functionals span V^*, because their normal vectors are the spanning roots and the positive definite form identifies V with its dual. Here is the missing faithfulness check on X. Any reflection changes $x\in X_{\mathbf{Q}}$ by a vector of V, so any word w does too. If w fixes V pointwise, it fixes all roots. Coroot compatibility in the root-datum axioms then makes its dual action fix every coroot. Consequently $\langle wx-x,\alpha^\vee\rangle=0$ for every root. But $wx-x\in V$, and the preceding restricted functionals span V^*, so $wx-x=0$. Thus the action on V is faithful on the full lattice, and W itself is finite. Now average the sum-of-squares form on all of $X_{\mathbf{R}}$ over this finite W. The same eigenspace calculation gives (1.2) for every $x\in X_{\mathbf{R}}$, not only for $v\in V$. We use this full-lattice invariant form from now on. Set

$$Z_{\mathbf{Q}}=\{x\in X_{\mathbf{Q}}:\langle x,\alpha^\vee\rangle=0\text{ for every }\alpha\}.$$

The last formula identifies $Z_{\mathbf{Q}}$ with the orthogonal complement of V in $X_{\mathbf{Q}}$ for this rational positive definite form. To see the decomposition over $\mathbf{Q}$ explicitly, choose a rational basis of V and solve its Gram linear equations for the projection of $x\in X_{\mathbf{Q}}$. The Gram matrix is invertible: a kernel vector would have squared norm zero. Gaussian elimination over $\mathbf{Q}$ gives a unique rational solution, whose subtraction from x is in $Z_{\mathbf{Q}}$. Thus $X_{\mathbf{Q}}=V\oplus Z_{\mathbf{Q}}$. Every reflection fixes the second summand pointwise. Equation (1.2) on V also shows that equal coroots have equal normal vectors and therefore proportional roots; their pairing-two identities force the proportionality constant to be one. Proportional coroots similarly have proportional normal vectors, hence proportional roots.

The **dual datum** interchanges X with $X^\vee$ and Φ with $\Phi^\vee$. Its axioms are the same identities read in the other direction. Reducedness is preserved, by (1.2). A datum is semisimple if $\mathbf{Q}\Phi=X_{\mathbf{Q}}$; in that case

$$\mathbf{Z}\Phi\subset X\subset\mathrm{Hom}(\mathbf{Z}\Phi^\vee,\mathbf{Z})$$

inside $X_{\mathbf{Q}}$. The left and right lattices are the root and weight lattices. The intermediate lattice is information that a picture of the roots alone cannot retain.

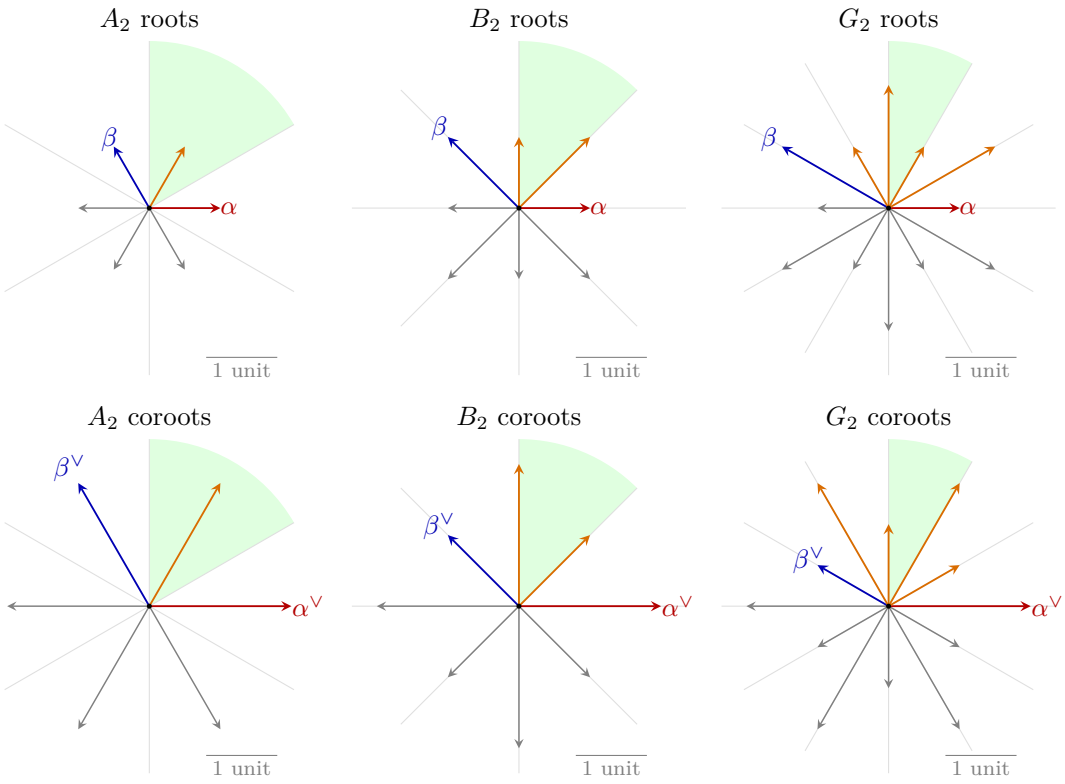


Figure 1. Roots and coroots in rank two. The top row uses the Euclidean coordinates $\alpha = (1, 0)$ and, respectively, $\beta = (-1/2, \sqrt{3}/2)$, $(-1, 1)$ and $(-3/2, \sqrt{3}/2)$. Red and blue mark the simple roots; orange marks the other positive roots; grey arrows are negative roots. The bottom row applies the metric identification $r^\vee = 2r/(r, r)$ from (1.2). The common scale makes the reversal of long and short lengths visible. Grey lines are reflecting hyperplanes, and the green sector consists of vectors pairing positively with both simple roots. This is a Euclidean realization of the rational spaces, not an identification of the integral character and cocharacter lattices. The definitions above and the reflection proof in Section 3 explain its mathematical content. The reproducible figure source and figure are original CC0 contributions.

2. The root datum of a reductive group

Let G be split connected reductive over a field k, and let T be a split maximal torus. Take

$$X = X^*(T), \qquad X^\vee = X_*(T), \qquad \mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha.$$

The pairing is composition of a character and a cocharacter. The preceding lesson proves that every root space is a line, that there are only the two roots $\pm\alpha$ on its rational line, and that the rank-one map $\varphi_\alpha : \mathrm{SL}_2 \to G$ provides an intrinsic coroot with pairing two.

Choose a nonzero vector in $\mathfrak{g}_\alpha$ and its linked dual in $\mathfrak{g}_{-\alpha}$. Write the corresponding root parametrizations as x_α and $x_{-\alpha}$. The element

$$n_\alpha = x_\alpha(1)x_{-\alpha}(-1)x_\alpha(1) \tag{2.1}$$

is the image of $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. It normalizes T. On the coroot torus it acts by inversion, and on the codimension-one torus killed by α it acts trivially. These statements follow in the rank-one group

and remain valid after its central quotient. On the rational character lattice they give precisely the reflection (1.1), since the two rational torus directions form an isogeny decomposition. Equality on the rational lattice is equality on the integral lattice, which is torsion-free. Thus

$$n_\alpha t n_\alpha^{-1} = t\,\alpha^\vee(\alpha(t)^{-1}). \tag{2.2}$$

Conjugation by any normalizer element permutes the root spaces and their root groups. The uniqueness of the rank-one coroot construction then permutes the coroots in the same way. Applied to (2.1), this proves both reflection axioms and their compatibility.

Theorem 2.1. $(X^*(T), \Phi, X_*(T), \Phi^\vee)$ is a reduced root datum. Its formation commutes with extension of the field.

Proof. Write $T \simeq \mathbf{G}_m^r$. The coefficient proof for diagonalizable groups in AG-GS, Theorem 2.2, identifies its characters with integer exponent vectors and its cocharacters with the dual vectors. Thus $X = X^*(T)$ and $X^\vee = X_*(T)$ are finite free of rank r, and their composition pairing is perfect. The finite-dimensional adjoint representation has finitely many weights by the full grading proof in RG01. The root-line reduction following RG03 Proposition 2.3 proves that every nonzero weight space is a line and that $\Phi \cap \mathbf{Q}\alpha = \{\alpha, -\alpha\}$. RG03 Theorem 7.1 supplies the intrinsic coroot $\beta = \alpha^\vee$ and the identity

$$\alpha \circ \beta(z) = z^2.$$

In particular $\langle \alpha, \beta \rangle = 2$ as an integer, in every characteristic, and $\beta \neq 0$. If Φ is empty, the reflection axioms are vacuous; the scalar-extension argument below still applies.

We make the torus decomposition in the reflection calculation precise. Write $\alpha = \nu\alpha_0$, where $\nu > 0$ and α_0 is a primitive exponent vector. Since

$$\nu\langle \alpha_0, \beta \rangle = 2, \qquad m := \langle \alpha_0, \beta \rangle = 2/\nu,$$

both ν and m are either one or two. Primitive means that the integer coordinates of α_0 have greatest common divisor one. The integer Euclidean algorithm gives integers b_i with $\sum_i a_i b_i = 1$: each division replaces a pair by its remainder pair, decreases the nonnegative remainder, and preserves its expression as a linear combination of the original coordinates. The cocharacter $\lambda_0(z) = (z^{b_1}, \ldots, z^{b_r})$ therefore satisfies $\alpha_0 \circ \lambda_0(z) = z$. Consequently

$$T \simeq T_0 \times \mathbf{G}_m, \quad T_0 = \ker \alpha_0, \qquad t \longmapsto \big(t\lambda_0(\alpha_0(t))^{-1}, \alpha_0(t)\big).$$

The inverse sends (u, z) to $u\lambda_0(z)$. The basis operations are integral and invertible. For a nonzero coordinate pair (a, b), put $d = \gcd(a, b) > 0$ and choose $ua + vb = d$ by that algorithm. Right multiplication of the row by

$$\begin{pmatrix} u & -b/d \\ v & a/d \end{pmatrix}$$

sends (a, b) to $(d, 0)$, and this matrix has determinant one. Repeat with the remaining coordinates, skipping zero pairs and adjusting a final sign. This sends the primitive row α_0 to $(1, 0, \ldots, 0)$. The corresponding monomial maps and their integer inverses are torus automorphisms by the coefficient classification, and identify T_0 with $\mathbf{G}_m^{r-1}$. In these coordinates $\ker \alpha = T_0 \times \mu_\nu$. For $\nu = 1$ or 2, the reduced identity component of μ_ν is the point, also when the characteristic is two. Hence the previously defined $T_\alpha = (\ker \alpha)^0_{\mathrm{red}}$ is exactly T_0.

Consider

$$p : T_\alpha \times \mathbf{G}_m \longrightarrow T, \qquad (u, z) \longmapsto u\beta(z).$$

In the preceding product coordinates, $\beta(z) = (c(z), z^m)$ for a cocharacter c of T_α. Changing the first source coordinate by $u \mapsto uc(z)$ identifies p with the identity on T_α times the power map $z \mapsto z^m$. This is finite free of positive rank m: over any test ring and any unit a, adjoining z with $z^m = a$ gives the free algebra with basis $1, z, \ldots, z^{m-1}$, and $z^{-1} = a^{-1}z^{m-1}$. It is therefore an fppf

cover. This proves the torus product statement with its full scheme structure; no invertibility of two is required.

Choose linked nonzero vectors in the two root lines. The rank-one homomorphism of RG03 Theorem 7.1 sends

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \text{to} \quad n_\alpha = x_\alpha(1)x_{-\alpha}(-1)x_\alpha(1).$$

Multiplying the three elementary matrices verifies this identity over every ring. Conjugation by the displayed matrix inverts $\mathrm{diag}(z, z^{-1})$, so $n_\alpha \beta(z) n_\alpha^{-1} = \beta(z^{-1})$. Each of the two root groups lies in $G_\alpha = C_G(T_\alpha)$, by its construction in RG03 Theorem 4.1; hence n_α centralizes T_α schematically. After the fppf cover p, write $t = u\beta(z)$. Then $\alpha(t) = z^2$ and

$$\begin{aligned} n_\alpha t n_\alpha^{-1} &= u\beta(z^{-1}), \\ t\beta(\alpha(t)^{-1}) &= u\beta(z)\beta(z^{-2}) = u\beta(z^{-1}). \end{aligned}$$

Equality of morphisms descends through this faithfully flat cover. Thus (2.2) holds over every test algebra. Its right side defines a torus automorphism: its value under α is $\alpha(t)^{-1}$, and applying it twice gives t. Therefore n_α normalizes T as a group scheme.

Applying an arbitrary character x to (2.2) gives exactly

$$x \longmapsto x - \langle x, \beta \rangle \alpha.$$

Applying (2.2) to a cocharacter y gives exactly

$$y \longmapsto y - \langle \alpha, y \rangle \beta.$$

These are identities on the integral lattices themselves. Substitution of $\langle \alpha, \beta \rangle = 2$ shows that both maps square to the identity and that their paired values equal $\langle x, y \rangle$. Conjugation transports adjoint weight spaces by this character automorphism and transports their root groups by the uniqueness in RG03 Theorem 4.1. It therefore permutes Φ. Transporting the full rank-one multiplication identity of RG03 Theorem 7.1, whose coroot and pairing are unique, gives

$$(s_\alpha \delta)^\vee = s_\alpha^\vee(\delta^\vee).$$

Thus the coroot image set is preserved and the correspondence respects both reflections.

We also verify that this correspondence is a bijection onto its image set. Put $V = \mathbf{Q}\Phi$. The group generated by the restricted reflections on V is finite because it permutes the finite spanning set Φ. Average any positive definite inner product on $V_{\mathbf{R}}$ over that finite group. Each restricted reflection fixes $\ker\langle\ , \alpha^\vee\rangle$, sends α to $-\alpha$, and is orthogonal. Decomposing a vector into its fixed part and its component on the normal line gives

$$\langle v, \alpha^\vee \rangle = \frac{2(v, \alpha)}{(\alpha, \alpha)}.$$

This argument uses only the root permutation and integer pairing already proved, so it does not assume injectivity of the coroot correspondence. If $\alpha^\vee = \delta^\vee$, their displayed normal vectors agree, making $\delta = c\alpha$. The two pairings with this common coroot are both two, so $2c = 2$ in $\mathbf{Q}$, giving $c = 1$. Thus $\delta = \alpha$. All the root-datum axioms now hold, and the already proved root-line assertion makes the datum reduced.

Finally let L/k be any field extension. The exponent-vector coefficient classification leaves both lattices and their pairing unchanged. The coaction weight projectors commute with scalar extension, and a nonzero weight space remains nonzero after faithfully flat extension of a field, so the root set and its lines extend with no added or lost weights. RG03 Theorems 4.1 and 7.1 prove scalar compatibility of the root maps, perfect pairings and unique coroots. Consequently all the constructed

data, including both reflection identities, are the scalar extensions of the original data. This proves every assertion. □

For GL_n, both lattices are $\mathbf{Z}^n$, with dual bases e_i and $e_i^\vee$. Conjugation by $\mathrm{diag}(t_1, \dots, t_n)$ sends the off-diagonal matrix unit E_{ij} to $t_i t_j^{-1} E_{ij}$ and fixes the diagonal units. The root subgroup is $I + uE_{ij}$, as is checked by multiplication. The embedded i, j block copy of SL_2 has the upper and lower parameters E_{ij}, E_{ji}; its matrix factorization from RG03 Theorem 7.1 and coroot uniqueness make its diagonal cocharacter the intrinsic coroot. Thus the roots and coroots are

$$\alpha_{ij} = e_i - e_j, \qquad \alpha_{ij}^\vee = e_i^\vee - e_j^\vee \quad (i \neq j). \tag{2.3}$$

For SL_2, write $T = \{\mathrm{diag}(t, t^{-1})\}$. In the resulting copies of $\mathbf{Z}$, the positive root is 2 and its coroot is 1. For PGL_2, use $t \mapsto \mathrm{diag}(t, 1)$ modulo scalars: the positive root is 1 and its coroot is 2. These two data are dual. Their root systems have the same two-point shape, but their integral data distinguish the groups even in characteristic two.

3. Reflections and chambers

In $X_{\mathbf{R}}^\vee$, remove the hyperplanes $\langle \alpha, \lambda \rangle = 0$. The connected components are **Weyl chambers**. They are open convex cones, possibly with an unrestricted central factor. A chamber determines the positive system

$$\Phi_\lambda^+ = \{\alpha : \langle \alpha, \lambda \rangle > 0\}. \tag{3.1}$$

Conversely its signs determine the chamber: a prescribed choice of strict signs is an intersection of open half-spaces, hence is convex. Such a set is connected: a segment between points in different parts of a supposed open separation would induce an open separation of $[0, 1]$. If its first part contains zero, openness and the supremum of the points up to which the whole interval stays in that part contradict either possible part containing the supremum. Thus an interval has no such separation. The hyperplanes are defined by integer linear forms on the cocharacter lattice. The needed rational approximation has this elementary coordinate construction: for each $\varepsilon > 0$ choose a positive integer n with $1/n < \varepsilon$ and, for a real coordinate a, an integer m with $m \leq na < m + 1$. These real-order facts follow directly from completeness. If the positive integers were bounded with supremum b, an integer $n > b - 1$ would give $n + 1 > b$, a contradiction. This gives the required n and integer bounds for na. Well ordering of the positive integers after shifting past a lower bound gives the least integer strictly above na; subtracting one gives the desired m. Then m/n differs from a by less than ε. Apply this in each of finitely many coordinates with errors small enough for the finitely many strict linear inequalities, and clear the positive common denominator. Thus every chamber contains an integral cocharacter. The central factor imposes no additional inequality. If Φ is empty there is one chamber, no simple root, and the reflection group is the identity; all assertions below then have their empty-set interpretation.

We first prove transitivity on chambers. Choose interior points p, q of two prescribed chambers so that their straight segment meets no intersection of distinct root hyperplanes. Here is the required choice argument. For two nonproportional defining forms f, g, a simultaneous zero on the segment would imply

$$f(p)g(q) - g(p)f(q) = 0.$$

This is a nonzero polynomial in p, q, since f, g are independent. Finitely many such zero sets cannot fill the open product of the two chambers: their product is a nonzero polynomial, and a polynomial vanishing on an open box is zero. The latter assertion follows by induction on the number of variables, regarding it as a polynomial in the last variable and using that a nonzero one-variable

polynomial has at most its degree many roots. Endpoints lie on no wall, so the chosen segment crosses finitely many hyperplanes, one at a time.

At a crossing point on just one hyperplane, take a sufficiently small ball meeting no other hyperplane. The corresponding orthogonal reflection fixes the crossing point, reverses the two half-balls, and permutes the full hyperplane arrangement by the root-datum axioms. It therefore interchanges the two adjacent chambers. The successive crossed-wall reflections carry the first chamber to the last, proving transitivity.

Fix a chamber and its positive roots. A positive root is **simple** if it is not a sum of two positive roots. We establish the simple-root assertions directly, before using a rank-two classification. For nonparallel roots a, b with $(a, b) > 0$, put

$$A = \langle b, a^\vee \rangle = \frac{2(a,b)}{(a,a)}, \qquad B = \langle a, b^\vee \rangle = \frac{2(a,b)}{(b,b)}.$$

These are positive integers, by (1.2) and the lattice pairing, and $AB = 4(a,b)^2/((a,a)(b,b)) < 4$. Indeed the nonzero vector $b - (a,b)a/(a,a)$ has positive squared norm $(b,b) - (a,b)^2/(a,a)$, because a, b are independent. Hence their product is one, two or three, and at least one of A, B equals one. If $A = 1$, then $s_a(b) = b - a$ is a root; if $B = 1$, then $s_b(a) = a - b$ is a root. This proves the difference assertion without importing a classification.

Distinct simple roots have nonpositive inner product: otherwise the positive one of these differences decomposes one of them. They are linearly independent. A dependence would separate into an equality $v = \sum_i c_i a_i = \sum_j d_j b_j$ of nonnegative real combinations with disjoint supports. Their mutual inner product is nonpositive, while (v, v) is positive unless $v = 0$. The chamber functional is strictly positive on a nonempty positive combination, excluding $v = 0$. A dependence with only one sign is excluded by the same functional. Thus there is no nonzero dependence.

Every positive root is a nonnegative integral combination of simple roots. Indeed, decompose a nonsimple root into two positive roots and continue with each nonsimple summand. If h is the chamber functional and $c = \min_{\delta \in \Phi^+} h(\delta) > 0$, each step down a branch lowers its value by at least c. Every branch visits strictly decreasing values in the finite set $h(\Phi^+)$, so has bounded finite depth. A binary tree of bounded finite depth has at most $1 + 2 + \cdots + 2^d$ vertices, by induction on the depth. The decomposition is therefore finite. Its terminal summands are simple. Since all roots are positive or negative, the simple roots are consequently a basis of $\mathbf{R}\Phi$, and every root has integral coefficients all of the same sign in that basis.

For a simple root a, reflection changes only its coefficient in this expansion. A different positive root has a nonzero coefficient at another simple root: parallel lattice vectors have rational ratio, so reducedness excludes any other positive root on the line of a. The unchanged positive coefficient forces the reflected root to remain positive. Thus s_a permutes $\Phi^+ \setminus \{a\}$.

The chamber inequalities are now exactly positivity on the simple roots. Each simple inequality defines a wall: in the dual basis, take its coordinate zero and all other coordinates positive. Every other positive root has a positive value there, so this is an interior point of that wall. Conversely all other inequalities follow from the nonnegative expansions. Let H be generated by these simple reflections. The preceding segment argument shows inductively that H is transitive on chambers: a wall of the current chamber vC, for $v \in H$, is v applied to a simple wall, and its reflection is vs_av^{-1}. Every root hyperplane is a wall of some chamber, by choosing a point on it avoiding the other finitely many hyperplanes. Transporting that chamber to C shows that its root reflection lies in H. Here the conjugation identity $vs_av^{-1} = s_{va}$ follows directly from (1.1), the dual pairing and coroot compatibility. Hence the simple reflections generate the full reflection group.

We can now classify any two-dimensional subsystem entirely within these axioms. For a plane P spanned by roots, $\Phi_P = \Phi \cap P$ is finite, spans P, is reduced, and is stable under its root reflections. Restrict (1.2) to P; its Cartan pairings remain integers. Choose a functional nonzero on its finitely

many roots. The argument just proved supplies precisely two simple roots a, b. Set

$$k = -\langle b, a^\vee \rangle, l = -\langle a, b^\vee \rangle.$$

Both are nonnegative integers, their product is less than four, and one vanishes exactly when both vanish, since their common numerator is $-2(a, b)$. Interchanging a, b if necessary gives

$$(k, l) = (0, 0), (1, 1), (2, 1), (3, 1).$$

In the last two cases $(b, b)/(a, a) = k/l$, so a is the short root. On coefficient pairs the two reflections are exactly

$$s_a(n, m) = (-n + km, m), \qquad s_b(n, m) = (n, ln - m).$$

Every root of Φ_P is in the orbit of a or b under these two reflections. For a nonsimple positive root $\gamma = na + mb$, both coefficients are positive by reducedness. Since

$$(\gamma, \gamma) = n(\gamma, a) + m(\gamma, b) > 0,$$

one of its inner products with the simple roots is positive. Reflection in that simple root subtracts a positive integer multiple of it, preserves positivity by the simple-root permutation already proved, and strictly lowers the positive integral height $n + m$. Repetition ends at a simple root. Negative roots follow by sign.

Here is the full orbit calculation. For each row below, write R for the ordered positive-root list; the displayed lists give the image of each entry under the indicated reflection, in the same order:

$$\begin{array}{r|l}
A_1 \times A_1 & R = (a, b) \\
 & s_a R = (-a, b), \quad s_b R = (a, -b) \\
A_2 & R = (a, b, a + b) \\
 & s_a R = (-a, a + b, b), \quad s_b R = (a + b, -b, a) \\
B_2 & R = (a, b, a + b, 2a + b) \\
 & s_a R = (-a, 2a + b, a + b, b) \\
 & s_b R = (a + b, -b, a, 2a + b) \\
G_2 & R = (a, b, a + b, 2a + b, 3a + b, 3a + 2b) \\
 & s_a R = (-a, 3a + b, 2a + b, a + b, b, 3a + 2b) \\
 & s_b R = (a + b, -b, a, 2a + b, 3a + 2b, 3a + b).
\end{array} \tag{3.2}$$

These images follow by substituting (n, m) into the two displayed coefficient maps. Each set $R \cup (-R)$ is therefore stable under both reflections and contains a, b, so it contains their entire orbits. Conversely, all entries are reached: in A_2, $s_a b = a + b$; in B_2, $s_b a = a + b$ and $s_a b = 2a + b$; in G_2, use $s_b a = a + b$, $s_a(a + b) = 2a + b$, $s_a b = 3a + b$, and $s_b(3a + b) = 3a + 2b$. Negatives are in the same orbits because $-a = s_a a$, $-b = s_b b$ and the maps are linear. The descent just proved shows that these are exactly all roots, with no missing orbit.

The orders needed in the word proof follow directly from matrices. On coefficient columns, $s_a = \left(\begin{smallmatrix} -1 & k \\ 0 & 1 \end{smallmatrix}\right)$ and $s_b = \left(\begin{smallmatrix} 1 & 0 \\ l & -1 \end{smallmatrix}\right)$. Their product R has determinant one and trace $kl - 2$. Multiplying a general two-by-two matrix by itself verifies $R^2 - (kl - 2)R + I = 0$. If $kl = 0$, both k, l are zero and $R = -I$, of order two. If $kl = 1$, the identity is $R^2 + R + I = 0$, so $R^3 = I$ and $R \neq I$, of order three. If $kl = 2$, it gives $R^2 = -I$, of order four. If $kl = 3$, it gives $R^2 - R + I = 0$ and $R^3 = -I$; $R \neq I$ and $R^2 \neq I$ (the latter would force $R = 2I$ in that identity, contradicting determinant one). Its order is therefore six. Write these orders as $m = 2, 3, 4, 6$. The identity $s_a R s_a = R^{-1}$ reduces every word to R^j or $R^j s_a$. The m powers in each list are distinct, and determinant distinguishes the two lists, giving exactly $2m$ elements. Both reflections fix the orthogonal complement of their plane in the root space, so these orders hold there; faithfulness proved in Section 1 makes them the orders on the full lattice too. This proves the full rank-two and word calculations without a trigonometric or topological input.

Length and standard parabolic cosets

We first prove the reflection assertion entirely in the root space, before using any group normalizer. This also makes the length and word results available for every reduced root datum.

Lemma 3.A (a chamber has trivial stabilizer). The reflection group acts freely on chambers.

Proof. For an empty root set the group is the identity. Otherwise Section 3 has already proved that the simple reflections generate W and that a simple reflection sends every positive root other than its own simple root to a positive root. We prove the precise deletion calculation before using length or chamber freeness.

Let $w = s_{i_1} \cdots s_{i_k}$ be any word and suppose $w\alpha_j$ is negative. Apply its letters to α_j from right to left. At some step a positive root first becomes negative. If that step is s_{i_r}, the positive root before it must be α_{i_r} by the simple-root sign property. For $v = s_{i_{r+1}} \cdots s_{i_k}$ we therefore have $v\alpha_j = \alpha_{i_r}$. The reflection conjugation identity gives $vs_jv^{-1} = s_{i_r}$.

It follows by multiplication, on the full lattice, that $ws_j = s_{i_1} \cdots s_{i_{r-1}} v$: precisely the letter s_{i_r} is deleted from the displayed word for w. This calculation applies to every word, with no reduced-word theorem assumed.

Take a nonidentity w and an expression with the fewest possible letters, writing it as $w = vs_i$. Its prefix v also has a shortest expression; a shorter prefix would shorten w. If $v\alpha_i$ were negative, the deletion just proved would express $vs_i = w$ with two fewer letters, a contradiction. Thus $\gamma = v\alpha_i$ is positive, while $w^{-1}\gamma = -\alpha_i$ is negative.

A chamber determines all root signs. If $wC = C$, a positive root γ is positive on wC, which says $w^{-1}\gamma$ is positive on C. The preceding positive root with negative inverse image is therefore impossible. Hence a chamber stabilizer contains only the identity on the root space, and the full-lattice faithfulness of Section 1 makes it the identity in W. □

The cell dimensions below now follow from this independent chamber proof. No normalizer identification is used in the length calculation.

Proposition 3.1. Let $\ell(w)$ be the minimum number of simple reflections in an expression for w. Then $\ell(w) = |\Psi_w|$, where $\Psi_w = \{\alpha \in \Phi^+ : w^{-1}\alpha < 0\}$. For $J \subset \Delta$, each right coset of $W_J = \langle s_\alpha : \alpha \in J \rangle$ has a unique element w of minimum length. It is characterized by $w(\Phi_J^+) \subset \Phi^+$. For this w and every $z \in W_J$,

$$\Psi_{wz} = \Psi_w \sqcup w\Psi_z, \qquad \ell(wz) = \ell(w) + \ell(z).$$

Proof. Fix the positive chamber C. A word in simple reflections gives a gallery of adjacent chambers from C to wC: if the current chamber is vC, its wall corresponding to a simple root leads to $vs_\alpha C$. Every such gallery must cross each hyperplane separating C from wC. There are $|\Psi_w|$ of these, since the sign of a positive root on wC is the sign of $w^{-1}\alpha$ on C.

Conversely choose interior points of C and wC generically, so the straight segment between them never meets an intersection of distinct root hyperplanes. A hyperplane with endpoints on the same side is not crossed; each separating hyperplane is crossed exactly once. The segment therefore supplies a gallery with $|\Psi_w|$ steps. Write its successive chambers as vC by the preceding adjacent-wall rule. At the end, $vC = wC$; freeness from Lemma 3.A gives $v = w$. Thus the gallery supplies a word of that length and proves the length formula. The same wall calculation gives $\ell(ws_\alpha) = \ell(w)+1$ or $\ell(w) - 1$ according as $w\alpha$ is positive or negative.

Choose a minimum-length element w of a right coset, which exists because W is finite. The last calculation gives $w\alpha > 0$ for every $\alpha \in J$. Every root in Φ_J^+ is a nonnegative integral combination of those simple roots; its image is consequently positive. Thus $w(\Phi_J^+) \subset \Phi^+$.

Each generator of W_J changes only the coefficient of its own simple root in a root's simple-root expansion. A positive root outside Φ_J has a positive coefficient outside J, which remains unchanged.

Since all coefficients of a root have the same sign, every element of W_J preserves its positivity. The corresponding assertion holds for negative roots outside Φ_J. In particular $\Psi_z \subset \Phi_J^+$ for $z \in W_J$.

Now suppose w sends Φ_J^+ to positive roots. If $\alpha \in \Psi_w$, then $w^{-1}\alpha$ is negative and outside the Levi subsystem: a negative Levi root would have negative image under w, whereas α is positive. Applying z^{-1} therefore leaves it negative. If $\alpha \notin \Psi_w$ is positive, then $w^{-1}\alpha$ is positive, and $z^{-1}w^{-1}\alpha$ is negative precisely when $w^{-1}\alpha \in \Psi_z$. These two cases prove the displayed disjoint inversion-root identity. Its cardinalities and the length formula prove length additivity. They also show that any element with the positive-Levi condition is a minimum in its right coset.

Finally, if two such minima satisfy $w' = wz$, applying additivity in both directions gives $\ell(w') = \ell(w) + \ell(z)$ and $\ell(w) = \ell(w') + \ell(z^{-1})$. Reversing a shortest word for z gives a word of the same length for z^{-1}, and reversing in the other direction gives the opposite bound. Hence $\ell(z^{-1}) = \ell(z)$, and we get $\ell(z) = 0$ and $z = 1$. This proves uniqueness. □

Relations between reflection words

The same length calculation determines all relations between simple reflections. This is the Coxeter presentation used in the pinned isomorphism argument of the next lesson. Its inputs are the rank-two root calculation (3.2), chamber freeness in Lemma 3.A, and the standard-parabolic coset calculation in Proposition 3.1. The proof below applies to every reduced root datum, including its central lattice directions.

For distinct simple roots α_i, α_j, put $c_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle \langle \alpha_j, \alpha_i^\vee \rangle$. The full rank-two calculation (3.2) and the two-by-two reflection matrices give $c_{ij} = 0, 1, 2, 3$ and exact product orders $m_{ij} = 2, 3, 4, 6$, respectively. Both reflections fix their common orthogonal complement; Section 1 makes the root-space action faithful on the entire lattice. The identity $s_i(s_is_j)s_i = (s_is_j)^{-1}$ then gives the m_{ij} powers $(s_is_j)^a$ and the m_{ij} elements $(s_is_j)^a s_i$ as all of W_{ij}, with distinctness proved by order and determinant as above.

After cancelling adjacent equal letters, words in two involutions alternate. Write $R = s_is_j$. Alternating words of length $2a$ are R^a and R^{-a}, while those of length $2a+1$ are $R^a s_i$ and $R^{-a-1}s_i$. Comparing these exponents modulo m_{ij} shows that all alternating words of length less than m_{ij} are distinct and reduced. The two alternating words of length m_{ij} agree and give the unique longest element. These words exhaust the $2m_{ij}$ elements: there is one empty word, two words at each length $1, \dots, m_{ij} - 1$, and one element at length m_{ij}.

A **braid move** replaces either alternating block of length m_{ij} by the other. With the involution relations, it is equivalent to $(s_is_j)^{m_{ij}} = 1$.

Lemma 3.2 (two left descents). If both s_i and s_j shorten w on the left, there is a length-additive factorization $w = v_0u$, where v_0 is the longest element of W_{ij} and u has minimum length in $W_{ij}w$.

Proof. Invert the unique minimum representative assertion of Proposition 3.1 to obtain the left-coset version. It gives $w = vu$, $v \in W_{ij}$, with $u^{-1}(\Phi_{ij}^+) \subset \Phi^+$ and $\ell(vu) = \ell(v) + \ell(u)$. A positive root outside Φ_{ij} has a positive coefficient on some other simple root; neither s_i nor s_j changes that coefficient. Such a root remains positive under W_{ij}, so the ambient length of v equals its rank-two length.

For $k = i, j$, the root $v^{-1}\alpha_k$ belongs to Φ_{ij}, and u^{-1} preserves its sign. Thus $w^{-1}\alpha_k < 0$ implies $v^{-1}\alpha_k < 0$. Every positive root in Φ_{ij} is a nonnegative combination of α_i, α_j, so v^{-1} makes all of them negative. Its length is consequently the number of rank-two positive roots. This is the rank-two longest element just calculated. □

Lemma 3.3 (reduced words). Any two reduced words for the same element are connected by braid moves.

Proof. Induct on the length k. If the first letters agree, cancel them, apply induction to the two tails of length $k-1$, and restore the first letter. Otherwise the two first letters are $s_i \neq s_j$, so both

shorten w on the left. Lemma 3.2 gives $w = v_0 u$, with $\ell(w) = m_{ij} + \ell(u)$. Fix a reduced word for u. Prefix it with either alternating length-m_{ij} word for v_0. Both resulting words are reduced; one starts with s_i, the other with s_j. After cancelling the first s_i, induction connects the first original word to the first constructed word. The same induction with s_j treats the second original word. A single braid move connects the two constructed prefixes. Composing these moves proves the assertion. □

Theorem 3.4 (Coxeter presentation). For the Weyl group of the root datum under consideration,

$$W = \langle s_i \mid s_i^2 = 1,\ (s_i s_j)^{m_{ij}} = 1\ (i \neq j)\rangle.$$

Proof. The relations hold by the reflection and rank-two calculations, so the presented group surjects onto W. Reduce an arbitrary word by induction on its number of letters. First reduce its prefix to a reduced word for its value a. If the final letter s_i increases length, the word is reduced. If it decreases length, choose a reduced word A for as_i. Then As_i is reduced for a, by Proposition 3.1. Lemma 3.3 changes the reduced prefix to As_i by braid moves. The entire word becomes $As_i s_i$, and the involution relation cancels the last two letters. Thus the given relations turn every word into a reduced word for its actual value. A word representing 1 reduces to the empty word. The kernel is zero. □

The argument applies to the full character lattice, including its central summand: reflections fix that summand, and the faithful action on the root span in Section 1 determines their relations.

4. Borel subgroups are positive systems

We first work over an algebraically closed field. For a Borel B containing T, the entire Borel-intersection argument in RG02 Section 1 says that $B \cap C_G(T_\alpha)$ is a Borel of the rank-one centralizer and contains T. RG03 Proposition 2.3 identifies its adjoint quotient with PGL_2 and its two Borels containing the diagonal torus with the upper and lower triangular groups. Their inverse images are the two rank-one Borels. Thus B contains exactly one of $U_\alpha, U_{-\alpha}$. The torus grading of $\operatorname{Lie} B$ is a subgrading of $\operatorname{Lie} G$. Its zero part is $\mathfrak{t}$, by RG02 Theorem 2.1. Its intersections with each pair of root lines are precisely the Lie algebras of the just identified upper or lower rank-one Borel, since taking the Lie algebra commutes with the schematic intersection. Hence

$$\dim B = \dim T + \tfrac{1}{2}|\Phi|. \tag{4.1}$$

Choose an integral cocharacter λ in a chamber, whose existence was proved in Section 3. The entire grading and infinitesimal proof of RG01 Lemma 3.3 gives smooth closed subgroups, their limit map and their Lie algebras. In particular $L_G(\lambda) = C_G(\lambda)$ contains T and has Lie algebra exactly $\mathfrak{t}$, since every root has nonzero pairing with λ. RG02 Lemma 1.1 makes this centralizer connected. The inclusion $T \hookrightarrow L_G(\lambda)$ has invertible differential between smooth groups, so S02 Lemma 3.A makes it étale; as a closed monomorphism it is an open immersion. Its image is both open and closed in the connected target and contains the identity, so it is all the target. Thus

$$L_G(\lambda) = T, \qquad P_G(\lambda) = T \ltimes U_G(\lambda), \qquad \operatorname{Lie} U_G(\lambda) = \bigoplus_{\langle\alpha,\lambda\rangle>0} \mathfrak{g}_\alpha. \tag{4.2}$$

The unipotent group $U_G(\lambda)$ is connected by the same RG01 lemma. Thus $P_G(\lambda)$ is connected solvable and lies in a Borel. Its dimension is (4.1), so its inclusion into the containing smooth Borel has invertible differential. S02 Lemma 3.A and the étale-monomorphism argument in S04 Lemma A1.9 make this closed inclusion an open immersion. Its image is nonempty, open and closed in the connected Borel, hence is the entire Borel, as a scheme.

For a positive root, the canonical T-equivariance of RG03 Theorem 4.1 gives

$$\lambda(t)x_\alpha(u)\lambda(t)^{-1} = x_\alpha(t^{\langle\alpha,\lambda\rangle}u).$$

This extends to $t = 0$ with value the identity on every test algebra, proving $U_\alpha \subset U_G(\lambda)$. We now prove that the Borel itself depends only on the chamber. Let μ be another integral point in that chamber, order the common positive roots in any way, and form

$$m : T \times \prod_{\alpha\in\Phi^+} U_\alpha \longrightarrow P_G(\lambda)$$

by multiplication. At the identity its differential is the direct-sum map from $\mathfrak{t}$ and the distinct positive root lines onto $\operatorname{Lie} P_G(\lambda)$; it is an isomorphism. Both sides are smooth, so S02 Lemma 3.A makes m étale on a neighbourhood of that point. Its image there is a nonempty open Ω by the complete étale-openness proof in S04 Lemma D2.3. Every factor also lies in $P_G(\mu)$. The defining ideal of $P_G(\mu)$ therefore vanishes on Ω: this can be tested on the surjective étale map to Ω, which is faithfully flat. Here $P_G(\lambda)$ is reduced and irreducible by the complete component argument in S08 Theorem G.4.2. The ideal therefore vanishes on all of it, since regular functions on an integral scheme inject into their values on a dense open. Thus $P_G(\lambda) \subset P_G(\mu)$ as closed schemes. Interchanging the two gives equality. Different chambers give different root summands in (4.2), so give different Borels.

Every Borel containing T is obtained. RG02 Section 1 proves that $N_G(T)/C_G(T)$ acts transitively on the Borels containing T, using conjugacy of maximal tori within a Borel and the full Borel normalizer theorem. If n carries one constructed Borel to B, conjugation carries its extended orbit limits to those for $n\lambda n^{-1}$, so

$$B = P_G(n\lambda n^{-1}).$$

The lattice action of n permutes the root hyperplanes, hence this is again a chamber cocharacter. This proves the correspondence over the algebraically closed field.

Over an arbitrary field k with split T, the integral cocharacter and the grading ideals defining $P_G(\lambda)$ are defined over k, and RG01 Lemma 3.3 proves their scalar-extension compatibility. Their geometric fibres are the Borels just described. Equality of their closed ideals can be checked after the faithfully flat extension $k \subset \bar{k}$. Hence the correspondence is defined over k, and each geometric Borel containing T descends to the indicated constructed subgroup.

Theorem 4.1. For a split reductive group over any field, Borels containing its split maximal torus correspond bijectively to positive systems. The normalizer Weyl group equals the reflection Weyl group and acts simply transitively on these systems.

Proof. The full preceding construction proves the correspondence and its compatibility with conjugation. Work geometrically first. RG02 Theorem 2.1 gives $C_G(T) = T$, so conjugation embeds $N_G(T)/T$ faithfully into the automorphisms of the full character lattice. RG02 Section 1 identifies its action on the Borels containing T as simply transitive: a stabilizer is $N_B(T) = C_B(T) = T$, and the torus-conjugacy argument gives transitivity.

Theorem 2.1 supplies, for each root, the element n_α of (2.1) with precisely the integral reflection action. Its coset in $N_G(T)/T$ is unique for that action, by the centralizer equality. Consequently the subgroup generated by these cosets identifies with the reflection group. Section 3 proves that this subgroup is transitive on chambers. If $v \in N_G(T)/T$, choose a reflection-group element w carrying a fixed chamber to v's chamber. Then $w^{-1}v$ stabilizes it, so is the identity. Thus every coset is in the reflection group, and its chamber action is free.

Finally choose the linked root frames over k; each is a one-dimensional k-vector space and the rank-one pairing is perfect. The canonical root maps, the opposite linked frames and (2.1) are therefore defined over k. A chosen reflection word for any w gives $n_w \in N_G(T)(k)$. All geometric Weyl actions thus have these k-rational representatives, and the preceding Borel correspondence descends with them. This proves every assertion over the given field. □

Theorem 4.2 (the relative Weyl scheme). $W_G(T)$ is represented by a finite étale group scheme over S.

Proof. Here G/S is reductive, T is a maximal S-torus, and $W_G(T)$ denotes the fppf sheaf quotient $N_G(T)/T$. A test point of the normalizer must preserve the entire subgroup T after that base change. We construct this quotient and its scheme structure over the original base.

Work on an étale cover splitting $T = D_S(X)$. On an affine neighbourhood, a finite generating set of the finite locally free module $\operatorname{Lie} G$ has coactions involving only finitely many characters of X. The coefficient projectors of the grading split it into finite projective weight summands. Their ranks are locally constant. Shrinking about a chosen point makes the occurring weights constant; a rank-zero summand is zero. RG03 proves that each nonzero summand is a root line. Proposition 7.4 of AG-GS, *Diagonalizable groups*, represents $\operatorname{Hom}(\mathbf{G}_m, T)$ and $\operatorname{Aut}(T)$ by constant exponent-map schemes. The finitely many coroot maps are therefore locally constant too. On a common neighbourhood we have one fixed finite reduced root datum, by the geometric theorem and its constant lattice pairings. Choose a positive base and trivialize its finitely many simple root lines. The perfect opposite pairings give the linked negative frames.

For a simple root write $\beta = \alpha^\vee$, and use the canonical root maps of RG03 Theorems 4.1 and 7.1 to define

$$n_\alpha = x_\alpha(1)x_{-\alpha}(-1)x_\alpha(1).$$

The rank-one homomorphism sends the matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ to this element, so it inverts $\beta(\mathbf{G}_m)$. We give the full torus calculation over every test algebra. Factor the fixed integer character as $\alpha = \nu\alpha_0$, with α_0 primitive. The integer pairing $\langle \alpha, \beta \rangle = 2$ gives $\nu m = 2$, where $m = \langle \alpha_0, \beta \rangle$, so m, ν are one or two. The unimodular construction in the complete proof of Theorem 2.1 gives a cocharacter λ_0 with $\alpha_0\lambda_0(z) = z$ and the product $T = T_0 \times \mathbf{G}_m$, where $T_0 = \ker \alpha_0$.

In these coordinates $\beta(z) = (c(z), z^m)$, and

$$T_0 \times \mathbf{G}_m \longrightarrow T, \qquad (u, z) \longmapsto u\beta(z)$$

becomes the identity times $z \mapsto z^m$ after the source coordinate change $u \mapsto uc(z)$. For any unit a of any ring R, $R[z]/(z^m - a)$ is free of positive rank m, with basis $1, z, \ldots, z^{m-1}$, and z is a unit. This proves that the displayed map is an fppf cover, including in characteristic two. The canonical torus equivariance gives $\alpha|_{T_0} = 1$, hence both opposite root groups and n_α centralize T_0. Fppf locally write $t = u\beta(z)$. Then $\alpha(t) = z^2$, and the rank-one matrix identity gives

$$n_\alpha t n_\alpha^{-1} = u\beta(z^{-1}) = t\beta(\alpha(t)^{-1}).$$

Equality descends through that faithfully flat cover. The right-hand torus map is involutive, since it sends $\alpha(t)$ to its inverse. Thus n_α normalizes T schematically and has the integral reflection action on X. This argument includes all nilpotent test algebras.

Let $W \subset \operatorname{Aut}(X)$ be the finite reflection group proved in Sections 1 and 3. Choose one simple-reflection word for each w and multiply its representatives to obtain n_w; the chosen action is w. For any test scheme $S' \to S$, Proposition 7.4 identifies the torus action of $n \in N_G(T)(S')$ with a locally constant map to $\operatorname{Aut}(X)$. Its value at each geometric point lies in W, by Theorem 4.1. The preimage of the complement of W is an open and closed scheme with no geometric point, hence is empty. This factors the map through W on the entire test scheme. On its open and closed part labelled w, $n_w^{-1}n$ centralizes T on all tests; RG02 Theorem 2.2 identifies that full schematic centralizer with T. Conversely every $n_w t$ normalizes T, and its label determines w uniquely. We have proved mutually inverse functor maps

$$\coprod_{w \in W} T \xrightarrow{\sim} N_G(T), \qquad (w, t) \longmapsto n_w t, \qquad n \longmapsto (w, n_w^{-1}n).$$

Thus the normalizer is represented on this cover. Right multiplication by T is ordinary translation in each labelled copy, so its fppf quotient is exactly W_S. Its group multiplication is the usual

multiplication of lattice actions; any difference between chosen word representatives has identity action and lies in T.

On overlaps the quotients represent the same sheaf $N_G(T)/T$. Their unique comparison isomorphisms respect multiplication and satisfy the cocycle identity. The complete module descent and algebra/morphism descent proofs in S04 Lemmas A1.2 and A1.5 therefore descend their affine algebras and group maps. We spell out finiteness and étaleness of this descent. Over an affine base $\operatorname{Spec} A$, choose finitely many affine pieces of the étale cover whose open images cover it, using S04 Lemma D2.3 and quasi-compactness. Their disjoint union is $\operatorname{Spec} B$, with $A \to B$ faithfully flat. Its quotient algebra C is a finite product, on these pieces, of finite products of their base algebras; it is finite projective as a B-module.

Descend C to an A-algebra D by that module lemma. The dual C^* has the dual descent datum, so descends to a module E. Evaluation and coevaluation descend to $E \otimes_A D \to A$ and $A \to D \otimes_A E$. Their triangle identities hold after the faithfully flat cover and therefore hold over A. Write the image of 1 under coevaluation as the finite sum $\sum_i d_i \otimes e_i$. The triangle identity is

$$d = \sum_i d_i \operatorname{ev}(e_i, d).$$

It gives a split surjection $A^r \to D$, so D is finite projective. It also gives finite presentation as an algebra explicitly: choose finitely many linear generators for the kernel of this split surjection, express 1 and every product $d_i d_j$ in the d_i, and impose those linear, unit and product relations on $A[X_1, \dots, X_r]$. Reduction of monomials gives an inverse module map from D to this quotient, so these finitely many relations present D.

For formal étaleness, let $R \to R/J$ be any square-zero extension of A-algebras and $D \to R/J$ a given map. After the faithfully flat base change by B, a map from the product algebra C is exactly a finite family of orthogonal idempotents with sum one, on each base piece. Idempotents lift uniquely through a square-zero ideal. Indeed, for a lift e with $\delta = e^2 - e \in J$, $2e - 1$ is a unit since $(2e-1)^2 = 1 + 4\delta$, and

$$e' = e - (2e-1)^{-1}\delta$$

is idempotent. The difference between two idempotent lifts satisfies $(2e' - 1)v = 0$ with $v \in J$, so is zero. Products of distinct lifted idempotents reduce to zero and are idempotent, hence are zero by uniqueness; their sum similarly equals one. The resulting local algebra lift is therefore unique. Its lifts on the double overlap agree, so the same affine descent lemma descends it to $D \to R$. This proves formal étaleness. Finite presentation and the complete infinitesimal criterion in AG-FSE, *Infinitesimal lifting and invariance under thickenings*, Theorem 4.1, make $\operatorname{Spec} D \to \operatorname{Spec} A$ étale; finite projectivity makes it finite locally free. The affine constructions and their uniquely descended group operations agree on base overlaps. They give the claimed finite étale group scheme over the original arbitrary S. □

5. Coordinates on the unipotent radicals

We need actual scheme coordinates. The following complete graded argument works over an arbitrary base ring, including one with nilpotents.

Lemma 5.1. Suppose a smooth affine R-scheme Y of finite presentation has a section e and a $\mathbf{G}_m$-action which extends to $\mathbf{A}^1$ and sends $0 \times Y$ to e. Suppose the tangent module at e is free with positive weights $d_1, \dots, d_N$. A $\mathbf{G}_m$-equivariant map from $\mathbf{A}^N_R$, with these weights, to Y whose differential at zero is an isomorphism is an isomorphism.

Proof. Put $A = \Gamma(Y, \mathcal{O}_Y)$. The full torus coefficient grading in RG01 gives $A = \bigoplus_{d \in \mathbf{Z}} A_d$. Extension of the coaction from $R[t, t^{-1}]$ to $R[t]$ forces every negative-degree component to be zero: these monomials are linearly independent even over a ring with nilpotents. Evaluation at $t = 0$ is

the map factoring through e. For $f \in A_0$ it gives $f = e^*(f)$ in A, so $A_0 = R$, with the inclusion split by the section. Its augmentation ideal is $I = \bigoplus_{d>0} A_d$. The fixed subscheme is precisely this section, so equivariance sends the zero section to e and intertwines the two augmentation maps.

Replace a finite algebra generating list by all of its finitely many homogeneous components, and discard the degree-zero components, already in R. If none remain, $A = R$; the tangent isomorphism gives $N = 0$ when $R \neq 0$, and the assertion is immediate. For $R = 0$ both schemes are empty, also giving the assertion. Otherwise let $q > 0$ and D be the minimum and maximum of these positive generator degrees. For $F_h A = \bigoplus_{d \geq h} A_d$, the monomial calculation gives

$$I^n \subset F_{nq} A, \qquad F_{(n-1)D+1} A \subset I^n.$$

Indeed a product of n positive generators has degree at least nq, whereas a monomial of length less than n has degree at most $(n-1)D$. Homogeneous expressions in the chosen generators span each component, so these inclusions survive all algebra relations. The two filtrations are cofinal. Consequently

$$\widehat{A}_I = \prod_{d \geq 0} A_d,$$

with multiplication defined by the finite sums of component products in each degree. In particular A injects into its completion, and a homogeneous element of completed weight d is exactly an element of A_d. The polynomial algebra $P = R[x_1, \ldots, x_N]$, with $\deg x_i = d_i$ and augmentation J, has the same properties.

Write $\phi : A \to P$ for the given map. Its differential is invertible along the entire zero section. Both schemes are smooth, so the complete differential proof in S02 Lemma 3.A makes the morphism étale on an open neighbourhood V of that section. We give the full completion inverse. For $n = 1$, the zero section is a map $\operatorname{Spec}(A/I) = \operatorname{Spec} R \to V$ over Y. Inductively,

$$A/I^{n+1} \longrightarrow A/I^n$$

has square-zero kernel I^n/I^{n+1}. Formal étaleness lifts the preceding map uniquely to $\operatorname{Spec}(A/I^{n+1}) \to V$, over the natural map to Y. All these thickenings have the same underlying space as the section, so the chosen étale neighbourhood applies at every stage. The complete infinitesimal criterion in AG-FSE, *Infinitesimal lifting and invariance under thickenings*, Theorem 4.1 supplies this formal property. The compatible maps send every element of J into I/I^n at stage n and therefore factor through $P/J^n \to A/I^n$. Taking inverse limits gives $\widehat{\psi} : \widehat{P}_J \to \widehat{A}_I$.

Their being maps over Y gives $\widehat{\psi}\widehat{\phi} = 1$. To check the other composite, test on every P/J^n: the identity map and the composite are two lifts of the same zero section, over the same A-map, through the étale $V \to Y$. Uniqueness makes them equal. Thus the completions are mutually inverse. Equivariance follows from this same uniqueness after adjoining the universal Laurent variable for $\mathbf{G}_m$. The completed inverse therefore preserves each integer weight. By the displayed product description it carries P_d into A_d; the forward map does likewise. Restricting the two inverses to their direct sums gives inverse algebra maps $P \leftrightarrows A$. This proves the assertion over the original arbitrary R. □

Work locally where the split torus, the root datum and the finitely many root lines are trivialized, and choose an integral chamber cocharacter λ. RG01 Lemma 3.3 gives the smooth closed $U_G(\lambda)$, its contraction and its positive Lie summands. It also gives $L_G(\lambda)$. The inclusion $T \to L_G(\lambda)$ has invertible differential because no root has zero pairing. S02 Lemma 3.A and the étale-monomorphism argument of S04 Lemma A1.9 make it an open immersion. On each geometric fibre the full proof of Section 4 identifies this centralizer with T; hence the open image is all of $L_G(\lambda)$. Thus (4.2) holds over this base, with $P_G(\lambda) = T \ltimes U_G(\lambda)$.

For a supplied Borel subgroup scheme B containing T with this positive system, its Lie weight summands are exactly $\mathfrak{t}$ and the positive root lines. This follows on the original base from their geometric fibre identities: the inclusions of the finite projective weight modules have invertible

matrices on every fibre, hence are isomorphisms locally. Apply RG01 Lemma 3.3 also inside the smooth affine B. Its $P_B(\lambda)$ has the whole Lie algebra of B, so its closed inclusion is étale and open by the same differential and monomorphism proofs. On every geometric fibre its nonempty open-and-closed image is all of the connected Borel; hence $P_B(\lambda) = B$. Functoriality of the limit now gives a closed inclusion $B \subset P_G(\lambda)$. Their displayed Lie gradings agree, so this inclusion is again étale and open, and the full field correspondence of Section 4 makes it surjective on all geometric fibres. Its open image is therefore the whole target. We have proved $B = P_G(\lambda) = T \ltimes U^+$ as schemes over this base. This includes nilpotents and is the relative Borel identity used in Theorem 8.1.

For any ordering $\alpha_1, \ldots, \alpha_N$ of Φ^+, multiplication is equivariant for λ, on the source by the positive weights $\langle \alpha_i, \lambda \rangle$. Its differential at the identity is the direct sum of these root lines onto $\operatorname{Lie} U_G(\lambda)$. The complete Lemma 5.1 therefore proves, on these trivializations,

$$\prod_i U_{\alpha_i} \xrightarrow{\sim} U_G(\lambda). \tag{5.1}$$

These are the given multiplication maps, so their inverses agree on overlaps by uniqueness and descend. This proves (5.1) for root line bundles as well. Changing the ordering changes coordinates on the same $U_G(\lambda)$. Changing the cocharacter within the positive chamber gives the same subgroup: the product image of the same root groups is the whole subgroup in both cases. We denote it by U^+. Applying the same proof to $-\lambda$, the full open-cell immersion of RG01 Lemma 3.3 gives

$$U^- \times T \times U^+ \hookrightarrow G. \tag{5.2}$$

Each construction is defined by the canonical root maps, multiplication and grading, and thus commutes with every base change.

The inverse of (5.1) supplies root coordinates. For nonparallel roots a, b, a positive system containing both exists: the two independent rational vectors a, b extend to a rational basis, by the Gaussian basis calculation in S06 Section 1. The rational dual functional assigning one to each of them pairs positively with both. Perturb this functional within rational functionals, retaining these inequalities and avoiding the finitely many root hyperplanes by the polynomial and approximation calculations of Section 3, then clear denominators as in Section 3. In a reduced root system $b \neq \pm a$ means exactly nonparallel. In the resulting U^+, the commutator has the form

$$[x_a(u), x_b(v)] = \prod_{ia+jb \in \Phi,\ i,j>0} x_{ia+jb}(C_{ij} u^i v^j). \tag{5.3}$$

Here a fixed ordering is used. To prove it, take the coordinate belonging to any root γ. It is a polynomial in u, v, because (5.1) is an isomorphism of affine schemes. Equivariance for the whole split torus makes each surviving monomial have character γ. Setting $u = 0$ or $v = 0$ gives the identity commutator, so its polynomial is divisible by both variables, by coefficient comparison over the base ring. A term can therefore occur only with $\gamma = ia + jb$, $i, j > 0$. Independence of a, b makes the exponent pair unique. This proves every coordinate of (5.3) over all test algebras; Laurent character comparison includes nilpotents and every characteristic.

More generally let C be a subset of the real root space closed under addition, and put $\Psi = \Phi^+ \cap C$. Set every root coordinate outside Ψ equal to zero. This defines a smooth closed subscheme parametrized by the root lines in Ψ. It is a subgroup. Indeed the coordinate polynomials for multiplication and inversion are torus-equivariant and vanish at the identity. Each nonconstant monomial has target character equal to the sum of its input characters, with their positive integer multiplicities. If all those characters lie in C, so does their sum; hence every coordinate outside Ψ is zero after either operation. These are polynomial identities over the base, proving the subgroup assertion schematically.

It is independent of the ordering. The product in a second ordering has no coordinate outside Ψ in the first ordering, by the same weight calculation, and has invertible identity differential.

Both sides are contracted by λ, so Lemma 5.1 makes the resulting coordinate map an isomorphism after trivializing the lines. These isomorphisms descend. Write the resulting closed subgroup as U_Ψ. If the same root subset occurs in two positive systems, the same ordered multiplication of its canonical root groups is an isomorphism onto both images. They are consequently the same subgroup of G, independently of the ambient positive system. For

$$\Psi_w = \Phi^+ \cap w\Phi^-, \qquad \Theta_w = \Phi^+ \cap w\Phi^+, \tag{5.4}$$

the relevant additive sets are the intersections of two strict positive or negative half-spaces, so are closed under addition. The two root subsets partition Φ^+. Ordering all roots in Ψ_w before those in Θ_w in (5.1) proves the scheme product

$$U_{\Psi_w} \times U_{\Theta_w} \simeq U^+.$$

The factors need not commute; this statement specifies their multiplication order.

6. The Bruhat cells from a projective limit

We give the projective limit argument, including its smoothness and its scheme structures over the field of definition.

Lemma 6.1. Let a $\mathbf{G}_m$ act on a smooth projective variety with an equivariant projective embedding. Suppose its fixed points are isolated. The points whose limit at zero is a fixed point p form an affine locally closed scheme Y_p, contracted to p, and its tangent space at p is the positive-weight part of T_pY. An equivariant map from a positive-weight affine space to Y_p, with invertible differential at zero, is an isomorphism.

Proof. First take p rational over the field in use. Diagonalize the vector space of the given linear projective action and write the weights as m_i. A fixed point is represented by a line in a single weight space, say of weight m. Choose a coordinate of that weight nonzero at p, and normalize it to one. Its affine chart U is invariant under $\mathbf{G}_m$ and is smooth affine. The other coordinates z_i have relative weights $m_i - m$. The limit equals p precisely when

$$z_i = 0 \quad (m_i < m), \qquad z_i = z_i(p) \quad (m_i = m).$$

The remaining coordinates have positive weights and vanish at p. Intersect these equations with $Y \cap U$. They give an affine closed subscheme of this chart, hence an affine locally closed $Y_p \subset Y$, with nonnegative coordinate grading, degree-zero field and contraction to p.

This also represents the limit condition on arbitrary test algebras. Within the chart a Laurent orbit $z_i(t) = t^{m_i - m} z_i$ extends to zero exactly when all negative coefficients vanish, and its value at zero gives the displayed zero-weight equations; Laurent basis independence proves this over every ring. Conversely those equations give the polynomial extension. We justify that an orbit with limit p must lie in this chart, rather than assuming its initial point lies there. The chart complement is invariant and closed. On any geometric point of a test, an initial point in that complement would have its entire punctured-line orbit there, so its zero limit would also be there: a closed subset containing the punctured affine line contains zero. This contradicts the chosen chart containing p. Thus every underlying geometric point of the test maps into the chart; an open immersion contains the entire scheme map precisely when it contains all its underlying points. The initial point, and hence its punctured orbit, therefore factor through the chart. The extension's inverse image of its complement misses both $D(t)$ and $V(t)$, which cover the entire underlying affine line even over a nonreduced test. It is consequently empty. The extension also factors through the chart. The displayed equations now apply to that whole test map. This proves the intrinsic limit functor and coordinate independence.

We prove smoothness by full equivariant square-zero lifting. Let $C \twoheadrightarrow \bar{C} = C/J$, with $J^2 = 0$, and let $y_0 \in Y_p(\bar{C})$. Its polynomial orbit is an equivariant map $F_0 : \mathbf{A}^1_{\bar{C}} \to U$ with $F_0(0) = p$. Formal

smoothness of the affine U, by the complete AG-FSE infinitesimal criterion, supplies a scheme lift $F : \mathbf{A}^1_{\bar{C}} \to U$. Differences of lifts are derivations into $J[t]$. If

$$E = \Gamma(\mathbf{A}^1_{\bar{C}}, F_0^* T_U), \qquad M = E \otimes_{\bar{C}[t]} J[t],$$

this module of differences is M: the smooth cotangent module is finite projective, and duality identifies its homomorphisms to the square-zero ideal with the displayed tangent tensor. The calculation takes place also on $C[t, a, a^{-1}]$ and on the analogous two-parameter ring; these polynomial and Laurent extensions are flat, so their kernel ideals are exactly the corresponding extensions of J.

Evaluation $M \to T_p U \otimes J$ is surjective, with kernel $M_0 = E \otimes_{\bar{C}[t]} tJ[t]$, because the finite projective E preserves the exact polynomial evaluation sequence. Correct F by such a derivation to make $F(0) = p$. On lifts use the action

$$(aF)(t) = a \cdot F(a^{-1}t).$$

Its reduction fixes F_0. The difference $c(a) = (aF) - F$ is therefore a regular section with values in M_0, and

$$c(ab) = c(a) + ac(b).$$

This follows by subtracting F through the intermediate aF; differences add because $J^2 = 0$. The module M_0 is stable under this action and is a quasi-coherent $\mathbf{G}_m$-module over $\bar{C}$. The full coefficient-integral homotopy in AG-GS, *Diagonalizable groups*, Proposition 7.14 applies even with infinitely many weights and supplies $w \in M_0$ with $c(a) = w - aw$. The corrected map $F' = F + w$ is then equivariant, since

$$(aF') - F' = c(a) + aw - w = 0.$$

Its value at zero is still p. Evaluation at one gives a lift $y \in Y_p(C)$ by the universal coordinate condition already proved. Thus Y_p is formally smooth on every affine square-zero test, and it is of finite presentation by its finite chart equations. The same infinitesimal criterion makes it smooth.

For the tangent calculation, a first-order orbit at p is $\sum_n t^n v_n$, where v_n is the weight-n part of a tangent vector. Extension permits exactly $n \geq 0$, and value zero at the fixed section excludes $n = 0$. Hence $T_p Y_p = \bigoplus_{n>0} (T_p Y)_n$. Its positive-weight contraction and smoothness now meet all hypotheses of the complete Lemma 5.1, which proves the last assertion.

The equations, orbit functor and lifting calculations commute with field extension. For a nonrational closed isolated fixed point, put $K = \kappa(p)$. The fixed scheme is smooth by the entire fixed-scheme proof in AG-GS-05 Theorem 3.4. On a geometric fibre an isolated smooth point has local dimension and tangent dimension zero, by the regular local cotangent criterion; finite-projective differential base change therefore makes the cotangent rank zero near p. The full arbitrary-base criterion in S07 Lemma 5.3A makes that smooth neighbourhood étale. Its field calculation proves that a finite-type algebra with zero differentials is a finite product of finite separable fields; it includes the full residue-field and nilpotent arguments. Applied to a sufficiently small affine neighbourhood it gives K/k finite separable. In the same projective chart impose the ideal of this $\operatorname{Spec} K$ point in the weight-zero coordinates and set the negative coordinates to zero. The resulting degree-zero algebra is K. As a K-scheme it is exactly the preceding chart construction in Y_K with the fixed coordinate values of p. It is affine over K and a locally closed k-subscheme of Y; the smoothness and tangent conclusions just proved over K give the asserted stratum. Finite separability contributes no tangent directions, by the same zero-differential field calculation. Geometric fixed points are handled over their field of definition; in the last assertion the affine space and its zero point are over that same field. This proves the original geometric and field versions. □

Let B correspond to Φ^+, put $X = G/B$, and choose an integral chamber cocharacter λ. The full flat free-action quotient and base-change proof in S07 Corollary 8.3 gives the fppf B-torsor $G \to X$. RG03 Lemma 5.1 proves that this quotient is a smooth projective scheme over the field in use, with the whole projective and arbitrary-base smoothness arguments supplied there. Geometrically the complete flag proof in RG01 Lemma 2.1 realizes X in a linear G-projective space. We use

that embedding over an algebraic closure for the projective limit calculation; all cells also have the explicit charts over k constructed below.

The T-fixed cosets are precisely n_wB, $w \in W$. Geometrically a fixed coset corresponds to a Borel containing T, because its full stabilizer is the conjugate Borel. The earlier Borel normalizer theorem identifies these cosets with the Borels themselves, and Theorem 4.1 gives their simply transitive Weyl action. The representatives are all defined over k.

There are no further λ-fixed points. On the geometric fibre, the closed fixed scheme X^λ is smooth by AG-GS, *Diagonalizable groups*, Theorem 3.4 applied to the stable affine charts of that projective embedding. Its finitely many components are disjoint closed-and-open smooth components, by the regular-local-domain calculation used in RG03 Lemma 8.B. Each is integral. Its product with the split torus is integral too: on any affine domain chart its algebra becomes a Laurent polynomial domain, and the dense chart overlaps remain nonempty. Thus the action's map to the finite discrete component labels is constant on that product. Its identity value gives the original component. Since T commutes with λ, this proves that every component is T-stable. Every such component contains a T-fixed point. To see this directly, choose an integral cocharacter η separating the finitely many T-characters of the embedding, by the integer separating-cocharacter construction in RG03 Lemma 2.1. For any point of that closed projective component, take its least nonzero η-weight projective limit. Separation leaves a single T-character space, so its line is T-fixed and remains in the closed component.

At n_wB, the torsor is smooth. Indeed after its fppf trivialization it is projection with the smooth affine factor B; S04 Corollary A3.2 descends affineness and Lemma A1.3a descends smoothness on those affine charts. Its fibre through n_w is n_wB. A tangent vector on X lifts through this smooth morphism by the square-zero lifting property applied to the dual numbers; the kernel is exactly the tangent of that fibre. Right translation by n_w^{-1} identifies it with the Lie algebra of $n_wBn_w^{-1}$ inside $\mathfrak{g}$. This gives the full tangent quotient

$$T_{n_wB}X = \mathfrak{g}/\operatorname{Lie}(n_wBn_w^{-1}).$$

The toral summand disappears, and its root weights are $\Phi \setminus w\Phi^+$. None is zero on λ. The fixed tangent space is therefore zero. Smoothness makes a neighbourhood in X^λ zero-dimensional and étale over the geometric field; its points are open reduced points by the full field calculation in S07 Lemma 5.3A. The point n_wB is also closed. Connectedness consequently makes its fixed component exactly this point. Thus the entire fixed scheme is the finite reduced set $\{n_wB\}$; all of these points are k-rational.

We identify its attracting strata with charts defined over k, retaining their full scheme structures. The open immersion (5.2) is right B-stable and is $U^- \times B$. Invariant-open descent in S04 Lemma A1.6 and the product torsor identity in S07 Corollary 8.3 give an affine open $V_1 \simeq U^-$ in X. Its translate

$$V_w = n_wV_1 \simeq n_wU^-n_w^{-1}$$

is an affine T-stable chart about n_wB. The ordered root coordinates of this group have characters $w\Phi^-$; this follows from the canonical root equivariance and uniqueness in RG03 Theorem 4.1. The roots on which λ is positive are exactly Ψ_w. The coordinate argument of Lemma 6.1 therefore cuts its attracting subscheme out by setting every negative-weight coordinate to zero. By the complete closed-root-subgroup proof in Section 5, it is precisely U_{Ψ_w}.

It is also the global attracting subscheme. A point with limit n_wB must lie in V_w: otherwise its orbit lies in the invariant closed complement, and its limit would lie in that complement too. This argument holds on every geometric point of a scheme test; an open immersion contains the whole test map when it contains all its underlying points. Within the chart, the polynomial orbit criterion of Lemma 6.1 works on every test algebra. Thus

$$U_{\Psi_w} \xrightarrow{\sim} X^+_{n_wB}, \qquad u \longmapsto un_wB. \tag{6.1}$$

Its tangent map is exactly $\bigoplus_{\alpha\in\Psi_w}\mathfrak{g}_\alpha$, the positive part of the displayed tangent quotient. These identities and charts are defined over k and commute with field extension.

Every geometric point of X has a unique projective λ-limit: choose the least nonzero weight among its homogeneous coordinates, divide the orbit coordinates by that power of t, and evaluate at zero. The defining homogeneous equations of the closed projective variety still vanish on this polynomial orbit extension. Thus its limit is in X and is fixed. Separatedness makes the extension unique; equivalently the normalized least-weight line fixes the limit. Consequently (6.1) gives a finite disjoint locally closed partition of X.

These strata are exactly the B-orbits as schemes. In (5.4), every root of Θ_w is in $w\Phi^+$, so its full root group, as well as T, belongs to the stabilizer $n_wBn_w^{-1}$. The torus normalizes both coordinate subgroups and $U^+ = U_{\Psi_w}U_{\Theta_w}$ in the specified order. Hence B's orbit map at n_wB factors through U_{Ψ_w}, and its image is exactly (6.1), including on nonreduced tests. In particular its schematic stabilizer inside B is $U_{\Theta_w}\rtimes T$: the remaining U_{Ψ_w} coordinate must be the identity, by the injectivity of (6.1).

Pull the full right B-torsor $G\to X$ back to this stratum. The morphism $u\mapsto un_w$ is a section. It gives mutually inverse maps

$$U_{\Psi_w}\times B\xrightarrow{\sim} Bn_wB,\qquad (u,b)\longmapsto un_wb,\qquad g\longmapsto (u,(un_w)^{-1}g),\tag{6.2}$$

where u is the inverse image of gB under (6.1). The torsor fibre-product identity proves that the second coordinate lies in B on every test scheme. The displayed target is the locally closed inverse image of the stratum in G, so these are isomorphisms with that exact scheme structure.

Theorem 6.2 (Bruhat decomposition). For a split connected reductive group over any field k,

$$G=\coprod_{w\in W}Bn_wB\quad\text{as a locally closed stratification,}\qquad G(k)=\coprod_{w\in W}B(k)n_wB(k).\tag{6.3}$$

Moreover Bn_wB has dimension $\dim B+|\Psi_w|$, and its quotient by the right B-action is an affine space of dimension $|\Psi_w|$.

Proof. The full preceding construction gives (6.1), (6.2), and the disjoint geometric covering. These schemes and maps are defined over k. Membership in a locally closed k-subscheme can be tested after $k\subset\bar{k}$: the open condition descends by its underlying inverse image, and its closed equations descend by faithful scalar extension. Thus every $g\in G(k)$ has its coset in exactly one of the k-defined strata. The inverse of (6.1) gives $u\in U_{\Psi_w}(k)$, and the full torsor relation gives $(un_w)^{-1}g\in B(k)$. Conversely (6.2) places every such product in that stratum. Since $U_{\Psi_w}\subset B$, these are exactly $B(k)n_wB(k)$. This proves the point decomposition for every field.

By (5.1), U_{Ψ_w} is affine space of dimension $|\Psi_w|$. The right B-action in (6.2) is translation on its B-factor, so its fppf quotient is exactly that affine space, and its dimension is $\dim B+|\Psi_w|$. The closures of the B-orbits in X, and of the $B\times B$-orbits in G, are invariant under those actions: translation preserves the orbit and hence its closure. Each closure is therefore a union of the finitely many orbits already identified. This gives the frontier property of the stated stratification as well. □

The disjoint-union notation in (6.3) describes strata and points. It does not assert that the cells are open-and-closed components, or that a morphism from a general scheme lands in one cell globally. In SL_2 the two cells have dimensions two and three, and the latter is open dense.

7. Central quotients and their root coordinates

Theorem 7.1. Let G be reductive over S and $H\subset Z(G)$ a closed subgroup of multiplicative type of finite presentation. Its affine reductive quotient G/H, constructed in the preceding lesson,

Theorem 1.1, commutes with every base change. For a split maximal torus T its maximal torus is T/H, and its root groups are the images of those of G.

Proof. RG03 Theorem 1.1 supplies the affine reductive quotient $q : G \to Q$, its full H-torsor identity, finite presentation and arbitrary base change. We prove the additional torus and root-group assertions as scheme statements. By RG02 Theorem 2.2, $C_G(T) = T$. Centrality therefore puts H inside T on every test scheme.

Work on an étale affine cover on which $T = D_R(X)$ and $H = D_R(M)$. This simultaneous cover exists by the complete relative character theorem in AG-GS, *Diagonalizable groups*, Theorem 7.7. On a further open neighbourhood the homomorphism is given by one map $\rho : X \to M$: choose the images of finitely many generators of X, using the group-like coefficient calculation of Lemma 2.1 and Theorem 2.2 there. The closed embedding gives a surjection $R[X] \to R[M]$. Its image has precisely the exponent basis in $\rho(X)$, so for $R \neq 0$ linear independence forces $\rho(X) = M$. The empty affine chart has nothing to prove. Put $X' = \ker \rho$.

The subgroup X' is a finite-rank free abelian group. Here is the integer argument: project a subgroup of $\mathbf{Z}^r$ onto the first coordinate. If its image is $d\mathbf{Z} \neq 0$, choose an element with first coordinate d; subtracting its multiples splits the subgroup as its cyclic span plus a subgroup of $\mathbf{Z}^{r-1}$. If the image is zero, use only that smaller subgroup. Induction proves freeness and finite generation, without requiring X' to be saturated. Thus $T' = D_R(X')$ is a split torus.

The H-grading on $R[X]$ has degree m consisting of the exponents with $\rho(x) = m$; its degree-zero algebra is $R[X']$. Choose, for each m, one lift $x_m \in X$. Then its degree-m component is the free rank-one $R[X']$-module $e^{x_m}R[X']$, and multiplication between components is an isomorphism. Consequently $R[X]$ is faithfully flat over $R[X']$. The full torsor algebra map is

$$R[X] \otimes_{R[X']} R[X] \longrightarrow R[X] \otimes_R R[M], \qquad a \otimes c_m \longmapsto ac_m \otimes e^m.$$

Its inverse on the exponent basis sends $a \otimes e^m$ to $ae^{-x_m} \otimes e^{x_m}$. These formulas are mutually inverse; the forward map is an algebra map, so its bijective inverse is an algebra map too, independently of the lifts. The morphism $T \to T'$ is of finite presentation: both are finite Laurent polynomial algebras over R; add their finitely many defining Laurent relations and the images of the finitely many source generators to a relative presentation. It is therefore an fppf H-torsor and represents T/H.

Write $A = \Gamma(G, \mathcal{O}_G)$. The closed inclusion $T \subset G$ gives a surjective equivariant algebra map $A \to R[X]$. Exactness of taking degree zero, proved in AG-GS Theorem 3.1, gives a surjection

$$A^H \longrightarrow R[X'].$$

It embeds T' as a closed subgroup of $Q = \operatorname{Spec} A^H$. The resulting map $T \to G \times_Q T'$ is a morphism of H-torsors over T' and is an isomorphism. Indeed, after an fppf cover with a section of the first torsor, equivariance translates that section to a section of the second; both torsors become H, and the map is translation by that section. The inverse descends uniquely. This also proves the exact inverse-image assertion $q^{-1}(T') = T$.

Now work where the root datum is constant and its root lines are trivialized. Section 5 proves the relative open cell $G_0 = U^- \times T \times U^+$. Since $H \subset T$ is central, its right action changes only the middle factor. This open is saturated for the H-torsor. S04 Lemma A1.6 descends it to an open $Q_0 \subset Q$. The displayed product torsor $G_0 \to U^- \times T' \times U^+$ has the same quotient functor, so uniqueness gives

$$Q_0 = U^- \times T' \times U^+ \subset Q. \tag{7.1}$$

For any root α, choose a positive system containing it. Its Borel is a closed subgroup $B = T \ltimes U^+$,

as proved in Sections 4 and 5. The surjection $A \to \Gamma(B, \mathcal{O}_B)$ is H-equivariant, so exact invariants give a closed subgroup

$$B' = B/H = T' \ltimes U^+ \subset Q.$$

The product identifies this quotient because H again acts only on T. As for T', the morphism of H-torsors $B \to G \times_Q B'$ is an isomorphism, so $q^{-1}(B') = B$. In its scheme coordinates $T' \times \prod_{\beta>0} U_\beta$, the map from U_α sets the torus coordinate to the identity and every other root coordinate to zero. It is a closed immersion into B', hence into Q. Its inverse image under q is exactly $H \times U_\alpha$ in the corresponding $T \times U^+$ coordinates, by the same torsor identity. In particular $H \cap U_\alpha = 1$, including on nonreduced test schemes.

Centrality makes the restriction of α to H trivial: in a local root frame, conjugation multiplies the root coordinate by that restricted character, whereas centrality makes this multiplier one on the universal H-algebra. Hence $\alpha \in X'$. The image root group has exactly that character for conjugation by T'. This can be checked after the faithfully flat cover $T \to T'$, where it is the original identity. None of these characters becomes zero, since $X' \hookrightarrow X$ is injective.

At the identity, (7.1) now gives the whole Lie decomposition

$$\mathrm{Lie}(Q) = \mathrm{Lie}(T') \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha.$$

All the displayed root summands retain their one-dimensional nonzero T'-characters. Thus the invariant tangent submodule is precisely $\mathrm{Lie}(T')$, using the full character grading, rather than character differentials. On a geometric fibre, any torus A' containing T' commutes with it, so $\mathrm{Lie}(A')$ injects into this invariant part. The inclusion of T' gives the reverse inclusion of tangent spaces, so its differential is an isomorphism. S02 Lemma 3.A and the étale-monomorphism proof in S04 Lemma A1.9 make $T' \to A'$ an open immersion. An open subgroup is also closed: every other coset is open, so their union is its open complement. Connectedness of A' therefore makes its nonempty subgroup all of A'. This proves that T' is maximal on every geometric fibre, hence is a maximal S-torus.

There are no additional roots in Q, because the displayed Lie decomposition is exhaustive. Apply the uniqueness assertion of RG03 Theorem 4.1 to Q: each closed smooth image just constructed has the required character and identity differential on its root line, so it is exactly the canonical root group of Q. This proves the assertion about all root groups.

All character algebras, grading projectors, torsor identities, product maps and coordinate-zero immersions above commute with arbitrary base change. On overlaps the maps are the original quotient map restricted to T and to each canonical root group, so their identifications agree. The complete morphism descent proof in S04 Lemma A1.5 glues them. This proves the theorem over the original base without a smoothness assumption on H. □

8. Splittings and subgroup constructions in families

Over a scheme, a split maximal torus need not trivialize its root lines. A reductive group is **split with specified root datum** here when its maximal torus is identified with $D_S(X)$, its datum is the specified constant datum, and its root lines are trivializable. A **pinning** chooses a nowhere-vanishing section of each simple root line. These are the objects used in the next lesson.

Every reductive group acquires such data étale locally. RG02 Theorem 2.3 gives a maximal torus. A torus has an étale splitting cover by its definition in RG01; the equivalent fpqc definition is justified by the complete relative character theorem, AG-GS Theorem 7.7. Work on an affine member with $T = D_R(X)$. The coefficient grading in RG01 gives

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha$$

with only finitely many nonzero weight modules locally. Each is a finite projective direct summand. Its rank is locally constant: on a local basis chart it is a fixed free rank, by S01 Lemma 1.2. After restricting around any chosen point the finite weight list and all its ranks are therefore fixed. RG03 Theorem 4.1 identifies its nonzero summands as root lines. The root-system assertion and the chamber construction of Sections 2 and 3 give the finite constant root system and a positive system.

The coroots are homomorphisms $\mathbf{G}_m \to D_R(X)$. By the same group-like coefficient proof in AG-GS Lemma 2.1 and Theorem 2.2, their finitely many character images are fixed integers on a further open neighbourhood. There are only finitely many coroots, so one common neighbourhood makes the whole datum constant. Trivialize *all* the finitely many root lines on a further common open cover. Choose the simple frames there; these are the pinning. Choosing all lines also verifies the stated split condition before using any later pinned classification theorem. This argument uses finite projective modules and universal characters over R, including its nilpotents.

The full product proof in Section 5 then applies. Its local inverses agree on overlaps by uniqueness, so without any choice of frames the product of positive root line bundles is U^+, the supplied positive Borel is $T \ltimes U^+$, and (5.2) is an open immersion. Choosing frames identifies these line bundles with affine coordinates.

For an arbitrary cocharacter λ, RG01 Lemma 3.3 gives smooth closed subgroups $P_G(\lambda)$, $L_G(\lambda)$ and $U_G(\lambda)$, their exact Lie weight decompositions, and the limit homomorphism. Its section is the inclusion of $L_G(\lambda)$; on all test schemes its kernel is $U_G(\lambda)$. The explicit mutually inverse maps

$$L_G(\lambda) \times U_G(\lambda) \longrightarrow P_G(\lambda), \qquad (l, u) \longmapsto lu,$$

$$g \longmapsto \big(l(g),\, l(g)^{-1} g\big)$$

give $P_G(\lambda) = L_G(\lambda) \ltimes U_G(\lambda)$. Here $L_G(\lambda)$ is the fixed subgroup for conjugation by λ. On a geometric fibre, a nonzero cocharacter has torus image: in integer coordinates divide its vector by its greatest common divisor; the resulting primitive vector is a closed one-dimensional subtorus and the map onto it is a finite free power map. The zero cocharacter has trivial image. RG02 Theorem 2.1 makes the centralizer of this torus image connected reductive. Thus all geometric fibres of the already smooth affine finite-presentation $L_G(\lambda)$ are connected reductive, proving that it is reductive over the base. Its roots have zero pairing; those of $P_G(\lambda)$ have nonnegative pairing and those of $U_G(\lambda)$ positive pairing, directly from the proved Lie grading. Relative parabolic quotients will be constructed in the last lesson.

For the quotient of Theorem 7.1, the lattice formula is $X' = \ker(X \to X^*(H))$. Its root characters lie in X', as the universal centrality calculation in that proof shows. The quotient of the rank-one homomorphism in RG03 Theorem 7.1 sends the original coroot to $q_T \circ \alpha^\vee : \mathbf{G}_m \to T/H$. For every $\chi \in X'$,

$$\langle \chi, q_T \circ \alpha^\vee \rangle = \langle \chi, \alpha^\vee \rangle.$$

In particular its pairing with α is two. The quotient transports the full rank-one multiplication identity, and Theorem 7.1 identifies both image root groups with their canonical parametrizations. The all-test coroot uniqueness of RG03 Theorem 7.1 therefore identifies $q_T \circ \alpha^\vee$ with the intrinsic coroot of Q. Characters change from X to X', and cocharacters by this dual map; the root system stays the same. The free subgroup X' need not be saturated, so central quotients can change the integral lattice even when H is finite.

These constructions are local on the base and are compatible with base change and the canonical root maps. They make no assertion of one constant lattice on a disconnected base. For example, $\mathbf{G}_m$ on one component and $\mathbf{G}_m^2$ on another is a reductive group of different torus ranks. Constant-datum assertions apply componentwise or on the covers just constructed.

Bruhat cells over a base

The field theorem also gives actual cells over the integers; the following argument identifies their scheme structures before using their fibres.

Theorem 8.1 (relative Bruhat cells). Let G/S be a pinned split reductive group of constant reduced root datum. Fix its split torus T, positive Borel B, Weyl group W, and the pinning representatives n_w. Let $N = |\Phi^+|$, $r = \operatorname{rank} X^*(T)$, and $\Psi_w = \{\alpha \in \Phi^+ : w^{-1}\alpha \in -\Phi^+\}$. Multiplication gives locally closed immersions

$$U_{\Psi_w} \times B \longrightarrow G, \qquad (u, b) \longmapsto u n_w b,$$

whose images C_w form a finite, disjoint decomposition of G into locally closed subschemes. With the root frames and a basis of $X^*(T)$, they have coordinates

$$C_w \simeq \mathbf{A}_S^{\ell(w)} \times \mathbf{G}_{m,S}^r \times \mathbf{A}_S^N.$$

The construction commutes with arbitrary base change, including to nonreduced schemes. For $S = \operatorname{Spec} \mathbf{Z}$, each stratum is a product of a split torus and an affine space over the integers. This locally closed decomposition is a finite partition of the underlying scheme points; it does not identify G with their coproduct as schemes or partition $G(R)$ for an arbitrary ring R.

Proof. The ordered root-product theorem of Section 5, applied relatively as above, identifies U_{Ψ_w} with the product of its root lines. The set Ψ_w is closed under addition of roots. Conjugation by n_w^{-1} identifies it with a closed root-product subscheme of U^-: complete its root ordering by the remaining negative roots, and set those remaining coordinates to zero. This is a closed immersion over S, since the ordered product identifies the whole U^- with a product of root lines. After left translation by n_w^{-1}, the displayed map is the composite

$$n_w^{-1} U_{\Psi_w} n_w \times B \;\hookrightarrow\; U^- \times B \;\hookrightarrow\; G.$$

The second map is the relative open-cell immersion already proved. Thus the original map is a locally closed immersion, with precisely the stated product as its source and image. This argument is over the base, rather than an inference of an immersion from geometric fibres. Inversion applied to the cell indexed by w^{-1} also gives the opposite-order coordinates $B \times U_{\Psi_{w^{-1}}} \to C_w$, $(b, u) \mapsto b n_w u$. A different pinning representative of the same Weyl element differs by an element of T and only changes the torus coordinate. Thus one can equally use $U^+ \times T \times U_{\Psi_{w^{-1}}}$. The inversion convention in this order is $\Psi_{w^{-1}} = \{\alpha > 0 : w\alpha < 0\}$.

On each geometric fibre, Theorem 6.2 identifies these images with the Bruhat cells. Hence their underlying sets cover G and are pairwise disjoint: every point lies over a point of S, and a geometric point above it tests both assertions. The length formula of Section 3 gives $|\Psi_w| = \ell(w)$. Finally $B = T \ltimes U^+$ as a group and $T \times U^+$ as a scheme, with $U^+ \simeq \mathbf{A}_S^N$ after choosing the root frames. These identifications prove the coordinate formula. All maps, coordinate-zero immersions and representatives base change, proving the last assertion. □

For a finite field $\mathbf{F}_q$ this gives the useful count

$$|G(\mathbf{F}_q)| = (q-1)^r q^N \sum_{w \in W} q^{\ell(w)}.$$

For a torification of the integral scheme, partition each affine coordinate by zero versus invertible. On a cell with $d = N + \ell(w)$ affine coordinates, a subset I of invertible coordinates gives a locally closed torus $\mathbf{G}_m^{r+|I|}$, with every other coordinate zero. These finitely many pieces partition geometric points and points over every field. They do not make each point over a ring belong to exactly one such piece. Thus the construction supplies the relative cells and counting hypotheses used in the lessons on finite-field counting and torified varieties without a claim about general ring-valued Bruhat double cosets.

A global equation for the longest cell

The longest cell is already open. An integral equation for this open will also control denominators in its coordinates.

Proposition 8.2 (the normalized longest-cell equation). Let $G/\mathbb{Z}$ be a pinned split reductive group with constant root datum and split torus $T = D_{\mathbb{Z}}(X)$. Fix positive roots Φ^+, put $s = |\Phi^+|$, and choose integral root frames $E_\alpha = dx_\alpha(1)$ for all roots. Write

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathbb{Z} E_\alpha, \qquad V = \bigwedge^s \mathfrak{g}, \qquad \eta = \sum_{\alpha \in \Phi^+} \alpha \in X.$$

Let v_+ and v_- be the ordered wedges of the positive and negative root frames. Choose a basis of $\mathfrak{t}$ and let $\varphi_- : V \to \mathbb{Z}$ extract the v_--coefficient in the resulting wedge basis. If n_0 represents the longest Weyl element, write

$$(\bigwedge^s \mathrm{Ad})(n_0) v_+ = \epsilon v_-, \qquad \epsilon \in \{\pm 1\}.$$

Then the regular function

$$d(g) = \epsilon^{-1} \varphi_-((\bigwedge^s \mathrm{Ad})(g) v_+)$$

satisfies $d(n_0) = 1$, and the opposite-order longest-cell map

$$\theta : \mathbb{A}^{\Phi^+} \times T \times \mathbb{A}^{\Phi^+} \longrightarrow G, \qquad (a, t, b) \longmapsto \psi(a) t n_0 \psi(b)$$

is an isomorphism onto $D(d)$. This holds for every reduced root datum, including rootless tori and every isogeny type.

Proof. AG-RG-03 Theorem 4.1 supplies the root line bundles and their canonical parametrizations; the split integral frames give the displayed decomposition. For a finite free module with ordered basis $f_1, \ldots, f_m$, its exterior power has the increasing wedges as a basis, with the following full calculation. Multilinearity and alternating relations express every wedge as a sum of increasing wedges, killing repetitions and changing signs under adjacent exchanges. The sign rule follows by expanding the alternating relation with two copies of $u + v$. For each increasing index set, the determinant of the corresponding coordinate rows defines an alternating multilinear functional: in its permutation sum the terms cancel in pairs if two columns agree. It therefore respects the alternating relations and evaluates to one on that chosen basis wedge and zero on every other increasing wedge. These functionals prove independence. Wedging images defines the exterior power of a linear map, since it preserves all these relations; wedging composites verifies composition, and wedging the inverse verifies invertibility. Thus the exterior adjoint action used below is a representation on this finite free wedge module over every test ring and commutes with scalar extension. AG-RG-03 Theorem 7.1 supplies the linked negative frame, the rank-one homomorphism and its Weyl representative. For each simple root, the image of $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is the integral element n_α of (2.1), and (2.2) gives its reflection action on T. Chamber transitivity and freeness in Section 3 give a unique element w_0 sending the positive chamber to the negative chamber. It sends every positive root to a negative root. Proposition 3.1 gives its length s; any other element has fewer inversions

and therefore smaller length. This proves existence and uniqueness of the longest element being used. Choose one word in simple reflections for each w, as allowed by Section 3, and multiply the corresponding n_α. This produces $n_w \in N_G(T)(\mathbf{Z})$ with the required action. Conjugation carries each integral root line isomorphically to the reflected line. Its coefficient relative to two $\mathbf{Z}$-frames has an integral inverse, so is 1 or -1. These facts prove the signed-wedge assertion used below. Choose an integral cocharacter λ strictly positive on Φ^+. There are exactly s positive-weight basis vectors of $\mathfrak{g}$, all remaining vectors having weight zero or negative. Consequently the unique maximal-weight line of V is $\mathbf{Z}v_+$, and its unique minimal-weight line is $\mathbf{Z}v_-$. Their torus characters are η and $-\eta$.

For a representation R on a finite free torus-graded module M and a vector m of weight ν, write

$$R(x_\alpha(z))m = m + \sum_{j>0} z^j m_j.$$

Conjugation by the universal torus and comparison of Laurent characters give $m_j \in M_{\nu+j\alpha}$. This comparison is an integral polynomial identity; it does not differentiate characters or divide by a prime. Positive root groups therefore fix v_+, and preserve the coefficient extracted by φ_-. By the ordered root-product theorem, for $u \in U$ these identities give

$$R(u)v_+ = v_+, \qquad \varphi_- \circ R(u) = \varphi_-.$$

Take $R = \bigwedge^s \mathrm{Ad}$. Integral normalizer representatives permute the integral root lines by units, so $R(n_w)v_+$ is a signed wedge indexed by $w\Phi^+$. Its v_--coefficient is zero unless $w\Phi^+ = \Phi^-$, which is precisely $w = w_0$. In the opposite-order coordinates of AG-RG-04 Theorem 8.1, the preceding identities thus show

$$d(utn_w u') = 0 \quad (w \neq w_0), \qquad d(utn_0 u') = \eta(t)^{-1}.$$

These are identities on the coordinate schemes, and remain true after every base change. The last expression is a unit and equals one at n_0.

The relative open-cell immersion of AG-RG-04 (5.2), translated and inverted as in Theorem 8.1, makes θ an open immersion. Its image is the longest Bruhat stratum. The strata cover all scheme points, so the computed restrictions show that this image and $D(d)$ have exactly the same underlying open set. Open subschemes carry the restricted structure sheaf and are determined by that set. They are therefore equal as schemes, including after nonreduced base changes.

If Φ is empty, then $\operatorname{Lie} G = \operatorname{Lie} T$ over $\mathbf{Z}$. The closed torus inclusion is a map between smooth groups with invertible differential, hence is étale by S02 Lemma 3.A and an open immersion by S04 Lemma A1.9. It is also closed and contains the identity in every connected geometric fibre. Its open complement therefore has no geometric point, so it is empty. Thus $G = T$ as schemes. Set $V = \bigwedge^0 \mathfrak{g} = \mathbf{Z}$ and $d = 1$; the asserted map is then the torus isomorphism. This proves the rootless case too. □

The sum η lies in the root lattice, hence in X, for every isogeny type. The proof requires neither individual fundamental weights in X, a perfect Killing form, nor a factorial coordinate ring or trivial Picard group. Orthogonal examples are understood as the smooth split reductive models of the stated root data; reductivity of every geometric fibre is among the hypotheses.

For PGL_2, take the class of $n_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Pull back the function to GL_2 and write $g = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$, $\delta = ae - bc$. Then

$$gE_{12}g^{-1} = \delta^{-1} \begin{pmatrix} -ac & a^2 \\ -c^2 & ac \end{pmatrix}, \qquad d|_{\mathrm{GL}_2} = c^2/\delta.$$

Here $\epsilon = -1$. The formula is scalar invariant and is the pullback of the already constructed regular function on PGL_2. Its value at n_0 is one. The boundary can have multiplicity two in this equation;

a principal open does not require a reduced boundary equation. This calculation is a pullback formula: it does not assert that every $\mathrm{PGL}_2(A)$-point has a $\mathrm{GL}_2(A)$-lift.

The exterior-power construction is due to Claude Chevalley, *Certains schémas de groupes semi-simples*, Section 4, Proposition 1, pp. 226–229. The proof here gives its normalized longest-cell form for every reduced root datum. Its integral representative input is the preceding rank-one construction, AG-RG-03 Theorem 7.1, together with (2.1)–(2.2) and the reflection words in Section 3 of this lesson.

9. Exercises

Exercise 9.1 (easy). Write the complete root datum of GL_n and identify its central rational direction.

Exercise 9.2 (medium). Verify integrally that the data of SL_2 and PGL_2 are dual. Explain why identifying their rational root systems does not identify their groups.

Exercise 9.3 (medium). Show that the Weyl group of GL_n is S_n, generated by its root reflections.

Exercise 9.4 (hard). Prove $\mathrm{GL}_n(k) = \coprod_{w \in S_n} B(k) w B(k)$ by upper triangular row and column operations, where B is upper triangular. Prove uniqueness of the permutation.

Exercise 9.5 (families). Let L be a nontrivial line bundle. For $G = \mathrm{SL}(\mathcal{O} \oplus L)$, describe the positive unipotent radical and decide whether the displayed diagonal torus has a global pinning.

Solutions

Solution 9.1

The lattices and roots are (2.3), with $\langle e_i, e_j^\vee \rangle = \delta_{ij}$. Reflection in $e_i - e_j$ exchanges coordinates i, j. The root span is $\{(a_i) : \sum a_i = 0\}$, while its orthogonal central direction is $\mathbf{Q}(1, \dots, 1)$. The centre has character group $\mathbf{Z}^n / \mathbf{Z}\Phi \simeq \mathbf{Z}$, the quotient map being summation. For $n = 1$ the root set is empty and the entire lattice is central.

Solution 9.2

In the coordinates of Section 2, the pairs (positive root, positive coroot) are $(2, 1)$ and $(1, 2)$ in dual copies of $\mathbf{Z}$. Interchanging lattices interchanges these pairs. The reflection is multiplication by -1 in both. The quotient of the character lattice by the root lattice is $\mathbf{Z}/2$ for the first and zero for the second, so their centres are μ_2 and the trivial group. This distinction holds scheme-theoretically in characteristic two.

Solution 9.3

Formula (1.1) for $e_i - e_j$ sends e_i to e_j, e_j to e_i, and fixes the other basis vectors. Thus its reflection is the transposition (ij). Every permutation is a product of disjoint cycles: start from a moved index, follow its successive images until it returns (injectivity rules out any earlier repetition), remove that finite orbit, and repeat on the remaining indices. With rightmost factors acting first, each cycle satisfies $(i_1\ i_2\ \cdots\ i_m) = (i_1\ i_m) \cdots (i_1\ i_3)(i_1\ i_2)$, verified by applying it to each of the listed indices; all other indices are fixed. Thus transpositions generate every permutation. Conversely all products of these reflections are coordinate permutations. The lattice representation is faithful, so the reflection group is exactly S_n; Theorem 4.1 identifies it with the normalizer quotient.

Solution 9.4

In the bottom row choose its leftmost nonzero entry, in column j. Scale that row to make the entry one. Add multiples of column j to columns to its right, making the other bottom-row entries zero; these are permitted right upper triangular operations. Entries to its left were already zero. Add multiples of the bottom row to higher rows to clear column j; these are permitted left upper triangular operations. Delete the bottom row and column j. The remaining square matrix is invertible, by expansion of the determinant along the cleared row. Repeat with its bottom row, retaining the original order of the surviving columns. Operations on these rows and columns extend to upper triangular operations on the original matrix and preserve every previously cleared pivot. Induction gives one nonzero pivot in every row and column, all scaled to one, hence a permutation matrix. This proves existence over every field.

For uniqueness consider, for every i, j, the rank r_{ij} of the submatrix in rows $i, \dots, n$ and columns $1, \dots, j$. Left upper triangular operations mix these bottom rows only with one another, and right upper triangular operations mix these first columns only with one another. Hence all r_{ij} are invariant. In a permutation matrix they count pivots in that rectangle. The difference

$$r_{ij} - r_{i+1,j} - r_{i,j-1} + r_{i+1,j-1}$$

is one exactly at a pivot (i, j) and zero otherwise, with empty boundary rectangles having rank zero. These invariants recover the permutation uniquely. The double cosets are therefore disjoint.

Solution 9.5

Choose the upper root as positive. Its group is the additive line bundle $\mathbf{V}(L^{-1})$, and the triangular subgroup is $T \ltimes \mathbf{V}(L^{-1})$. A pinning would choose a nowhere-vanishing section of L^{-1} and therefore trivialize L. Hence this pair has no global pinning, although it has one on every open set trivializing L. Its relative big cell is $\mathbf{V}(L) \times T \times \mathbf{V}(L^{-1})$.

The course prerequisite guide records the exact supporting statements, their full proof routes, and the hypotheses needed in their applications.

References and prerequisites

The root datum terminology is due to Demazure; freely accessible comparison is SGA 3, Exposé XXI, 13 October 2024 re-edition. For the group constructions see Conrad's author manuscript, *Reductive group schemes* and Milne's freely accessible *Algebraic Groups*, version 2.00 (2015). The reflection, word, coordinate and cell arguments are written here; these sources do not replace a programme proof.

The general fibre criteria used in Section 7 belong to the programme's morphism theory. The geometric homogeneous quotient used in Section 6 is the field quotient construction introduced in Quotients and torsors, together with the self-normalizing projective-quotient construction in the preceding lesson. Their precise prerequisite locators and proof states are recorded in the course dependency list.